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Canadian evidence on ten years of universal preschool policies: The good and the bad☆

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HIGHLIGHTS

- We provide a comprehensive review of the impact of the Québec universal childcare reform.
- The reform had long lasting positive effects on mothers` labour force participation.
- Childcare participation and hours in care increased and remained at high levels even 10 years after the reform implementation.
- School readiness measured using the PPVT did not improve.

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ABSTRACT

More than ten years ago, to increase mothers' participation in the labour market and to enhance child development, the province of Québec implemented a $5 per day universal childcare policy. This paper provides a comprehensive review of the impact of the program over more than 10 years after its implementation. A nonexperimental evaluation framework based on multiple pre- and posttreatment periods is used to estimate the policy effects. We find that the reform had important and lasting effects on the number of children aged 1–4 years old attending childcare and the numbers of hours they spend in daycare. For children aged 5 years old, we uncovered strong evidence that implementing full-day kindergarten alone was not enough to increase maternal labour force participation and weeks worked, but when combined with the low-fee daycare program it was, and these effects were also long lasting. Finally, our results on cognitive development suggest the reform did not improve school readiness and may even have had negative impacts on children from low-income families.

1. Introduction

In recent decades policies geared towards families with preschool children have become a matter of serious public concern in the developed world. Family oriented policies – such as maternity and parental leaves, and universal and targeted childcare – have appeared under different forms in numerous countries.[1] Each particular public policy assumes that subsidising parental leave and childcare services for children of working parents, and particularly working mothers, is in the public interest. This support for mothers in the labour market is seen by many to be crucial for the protection of children against

☆ The analysis is based on Statistics Canada's National Longitudinal Survey of Children and Youth (NLSCY) and Survey on Labour and Income Dynamics (SLID) restricted-access Micro Data Files, which contain anonymous data. All computations on these micro-data were prepared by the authors who assume responsibility for the use and interpretation of these data. The authors would like to thank Marie Connolly, Christopher Flinn, the referees and participants at the CEA, ESPE, ACFAS and SCSE conferences for many helpful comments. This research was funded by the Fonds québécois de la recherche sur la société et la culture.
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[1] In Europe, both parental leave and childcare services have been at centre stage. Since the 1990s, the European Parental Leave Directive (2009) sets mandatory minimum thresholds and binding guidelines for both the length and compensation of parental leave across member countries. Since 2002, the European Council sets out targets regarding childcare ('Barcelona targets of 2002'). The aim was that each country should provide childcare to at least 90% of children from age 3 years old up to mandatory school age, and to at least 33% of children younger than 3 years old, by 2010.

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poverty, in particular children of single mothers. Protecting children against poverty enhances cognitive development and health, not only during childhood but also in adulthood (e.g., Del Boca and Wetzels, 2007). From this perspective, the main policy rationale is not only to enhance the balance between the work place and family life, but also to contribute to child development and socioeconomic integration.

In the past decade, the number of economic studies evaluating the impact of expansions of childcare programs has grown considerably. Recent studies (reviewed below) have essentially evaluated the impact of country-specific daycare reforms on maternal labour force participation. To date, the results on labour force participation are mixed, but this may come as no surprise considering the wide variety of reforms studied and the contextual settings in which they take place. Other objectives of these programs, such as child development and childcare participation, are rarely studied.

In this study, we provide a comprehensive account of the impacts of a universal preschool program in the Canadian province of Québec. This large scale 'natural' policy experiment has unique characteristics compared to the diversity of policies studied in other countries. First, it provides universal coverage (regardless of parents' employment, marital status, or income) for all children aged 0–5 years old. Second, the fee parents pay is fixed ($5 per day at implementation, $7 since 2004). Third, it provides year-round services (261 weekdays) for a minimum of 10 hours per day. Several studies of this program were conducted shortly after its implementation in 1997 (e.g., Baker et al., 2008; Lefebvre and Merrigan, 2008; Lefebvre, Merrigan and Verstraete, 2009), but few have examined the long-term effects of the program (e.g., Kottenlenberg and Lehrer, 2013).

The unique contributions of this paper are that we first investigate the impact of the reform over 11 years compared to 5 or 6 years in previous work. Second, we allow our estimated impacts to vary by year to reflect the progressive increase in number of low-fee spaces. Third, we analyse the impact of the reform on a large number of outcomes within a common framework. Outcomes not yet covered in previous studies include the number of hours children spend in childcare (conditional on attending childcare or not), the number of weeks worked by the mother, and paternal labour force participation and weeks worked. A comprehensive review of this nature is rarely provided, yet it is crucial to understanding the specific and long lasting effects of a program. Fourth, we provide evidence by mothers' education level to assess the impact of the reform from an equity standpoint. Although the fee is the same for everyone, the postreform after-tax fee was almost identical to the prepolicy fee for low-income households, but was greatly reduced for higher income households. As a result, the incentive to work was larger for mothers having access to better-paid jobs. Fifth, we estimate the impact of the policy by child's age to account for the phase-in of the policy conditioned on children's ages. Finally, this paper provides general lessons regarding the Québec program.

We use Statistics Canada's National Longitudinal Survey of Children and Youth (NLSCY) for the estimation of the impacts of the policy, and we rely on a nonexperimental framework where the evolution of outcomes for Québec children and their parents are compared with those of similar children and parents in the Rest of Canada (RofC).

The results presented in this paper show that children's time in daily care that is conditional on attending daycare increased significantly under the program, and so did maternal labour force participation and weeks worked. We find that more educated mothers reacted more rapidly to the policy, and their children gained access to a larger proportion of the new subsidised childcare spaces. On paternal labour force participation and children's cognitive development, we find no significant effects when looking at the entire sample. Finally, when we break down the results by education level, we find a negative effect on child development in the later years of the program for children of less-educated

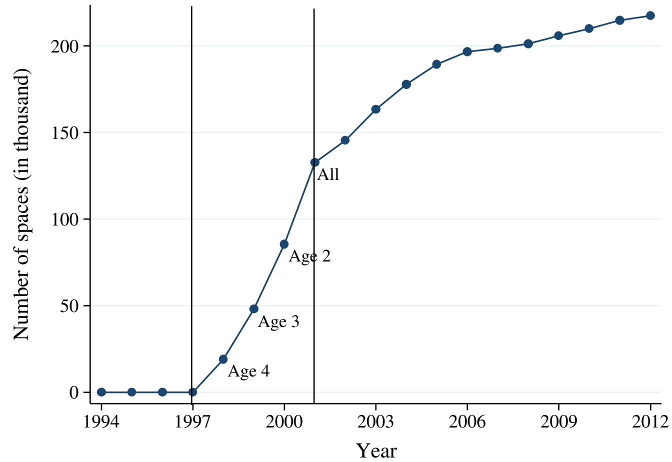

Fig. 1. Number of low-fee spaces. Note: Shows the evolution of the number of low-fee spaces between 1994 and 2012. The number of spaces is measured on March 31 of each year by the Direction générale des services de garde, Ministry of Families and Elders (MFA). The number of spaces between 1998 and 2000 is estimated according to the proportion of children by age group. The data can be accessed at www.mfa.gouv.qc.ca/fr/services-de-garde/portrait/places/Pages/index.aspx.

mothers. This is distressing as one of the main goals of the policy was to increase school readiness, in particular for children from low-income families.

The outline of the paper is as follows. Québec's early childhood education and care (ECEC) program is presented in Section 2, followed by a review of the evidence from prior research in Section 3. Section 4 presents the empirical strategy, while Section 5 briefly describes the data set. Econometric results on the impact of the daycare policy on hours of formal childcare and parental labour supply are presented in Section 6, and on cognitive test scores for children aged 4–5 years old in Section 7. Section 8 concludes by providing, to the extent possible, lessons for policymakers.

2. Québec's childcare reform

The Québec Early Childhood Education and Care (ECEC) program pursued two explicit objectives: (1) increase mothers' labour force participation (while balancing the needs of workplace and home), and (2) enhance child development and school readiness. We first discuss the implementation schedule of the policy, and then we describe the pre- and postreform environments.

On September 1, 1997, childcare facilities licensed and regulated[2] by Québec's Ministry of the Family and Elders (MFA) started offering spaces at the low fee of $5 per day per child. Between 1997 and 2000 the program was phased in, conditional on the age of the child. Starting September 30, 1997, children 4 years of age became eligible. They were followed by the 3-year-olds on September 1, 1998 and by the 2-year-olds on September 1, 1999. By September 1, 2000, all children aged 0–5 years old were eligible.[3] The number of spaces by age group for this period is not available, but clearly the number of spaces in low-fee daycare gradually increased between 1997 and 2000. As of September 2000, the number of low-fee spaces could accommodate only about 20% of Québec children. Fig. 1 shows the total number of low-fee spaces over time. Between 2000 and 2012, the number of spaces gradually

[2] Regulated childcare may take a variety of forms: not-for-profit daycare centres, for-profit daycare centres, and family-based daycare (caring for no more than 6 children).

[3] The exception is children age 5 years old (60 months) as of September 30.These children enter public school kindergarten. This is the first year of preprimary school for most children in Canada. Age eligibility is different in the other provinces (age 5 years old on December 31). Kindergarten is not compulsory, but almost all children are enrolled.

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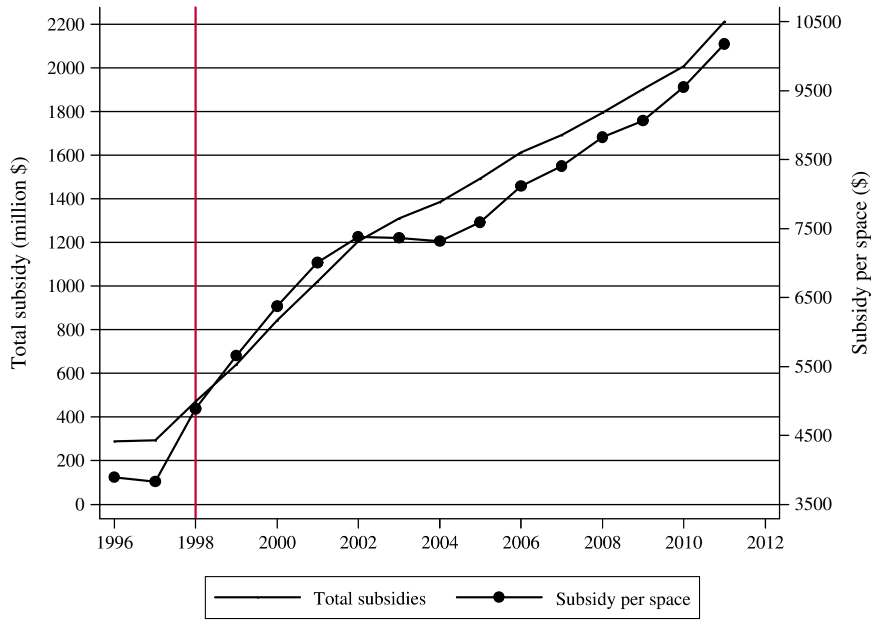

Fig. 2. Total public subsidies and subsidy per space. Note: The solid line shows the evolution of the total annual public subsidies allocated to the ECEC program in millions of dollars (left axis). The solid-dotted line shows the annual subsidy per space in dollars (right axis). The subsidy per space is the mean subsidy for both providers and families before 2000 and low-fee spaces since 2000. Our sources of data are the Ministry of Families and Elders (number of spaces) and the Treasury Board (total subsidies).

increased from 85,000 to 217,000 spaces and thereby released the capacity constraint of the childcare network. In our empirical strategy, we allow our estimated effects to vary by year to reflect the growing number of spaces over time.

The increasing number of spaces was paralleled by increasing costs to the government. Prior to 1997, the government provided small grants to regulated facilities to cover part of their fixed costs. It also helped low-income families with a childcare subsidy and a tax credit for eligible childcare expenses subject to certain limits. The tax credit varied from 75% for low-income households to 26% for high-income households. As of September 1997, subsidies to low-income households were gradually abolished, but the refundable tax credit remained available for families using daycare. The low-fee network was financed by both the government and the parents. The government paid a large fraction of the costs through a fixed amount per child while the parent paid the reduced fee of $5 per day. These childcare expenses were not eligible for the provincial tax credit. At the individual level, the prepolicy after-tax fee was almost identical to the postpolicy fee for low-income families, but it was greatly reduced for middle- to high-income families. For the government, the overall cost prior to the reform was very modest compared to the overall cost of the low-fee childcare network. During 1996–1997, governmental childcare subsidies, to both childcare facilities and low-income households, amounted to 288 million dollars. By 2011–2012, total governmental subsidies directly contributed to providers 2.2 billion dollars.

Fig. 2 shows the provincial government's total direct cost[4] and average cost per space for the childcare program from 1996 to 2012. In the first year of the policy, the mean subsidy per space was $3888. For fiscal year 2011–2012, the subsidy amounted to $10,210 per space. Two main reasons explain this dramatic increase. First, while operating costs have increased over the years, the fee paid by the parents increased only once over the period, from $5 to $7 per day in 2004. Second, childcare educators were able to negotiate much better working conditions over the years, and most even became unionised. The mean cost masks important differences by setting and age of children: not-for-profit centres receive the highest average subsidy per space ($13,235), followed by for-profit centres ($10,840), and family-based spaces ($8514). Furthermore, the subsidy is higher for children younger than 18 months as the children to educator ratio is lower.

Using administrative data, we cannot trace any such elaborate picture of the evolution of childcare services in the other provinces in Canada. However, Table A.1 provides the number of regulated spaces by province for 2001 and 2006, as well as the number of children receiving subsidies. In 2006, 38% of regulated daycare spaces across Canada were in Québec (200,005 versus 325,753 spaces in the other provinces) but fewer than one-fourth of the children lived in Québec. While 98% of the regulated spaces in Québec were totally subsidised, only 48% were totally or partially subsidised in the RofC. Table A.1 also stresses that Québec has a unique childcare regime compared to those existing in the other provinces in terms of provincial funding,[5] monthly or daily fees, and eligibility. When we discuss our results we provide further details on the environment at the time of the reform.

Concerning kindergarten children, although there are differences in kindergarten programs across Canada, the only major change in kindergarten policy over the period of our study occurred in Québec.[6] Prior to 1997, children who reached age 5 years old by September 30 were eligible to start half-day kindergarten (2 hours and 30 minutes). As of September 1998, full-day kindergarten was implemented in all public schools. Kindergarten is not compulsory,[7] but if a child is enrolled in a public school, he or she must attend class for the full school day and school week. Furthermore, schools were required to offer before- and after-school childcare for children age 5 years old or older for $5 per day ($7 since 2004).

[4] The direct costs abstract from the childcare refundable fiscal credit for unsubsidised daycare.

[5] In the other provinces, licensed childcare providers may receive one-time funding (for the expansion of spaces) or recurrent funding (for equipment, infrastructure, administration, and salaries).

[6] Two provinces (Nova Scotia and New Brunswick) offer full-time kindergarten. In the other provinces, it is half day. The province of Ontario offers half-day junior kindergarten for 4-year-olds.

[7] In Québec, as of September 1998, almost all 5-year-olds attended full-day kindergarten (98% compared to 85% before the policy change).

3. Prior research evidence on ECEC

Research on early childhood education and care has grown considerably over the past decade. In this section, we review some of the important contributions on the links between ECEC programs and the following: (1) maternal labour force participation and childcare use, and (2) children's cognitive development. In both cases, we focus first on studies of the Québec reform, and then we survey studies on reforms from other countries.

3.1. On maternal labour supply and childcare use

Earlier studies found that the Québec experiment produced an important increase in the labour supply of mothers of eligible children. Lefebvre and Merrigan (2008) used Statistics Canada's Survey of Labour and Income Dynamics (SLID) annual data from 1993 to 2002 to estimate the impacts of the policy on the labour supply of mothers with young children. Using the sample of all Canadian mothers with at least one child aged 1–5 years old, they found that the policy had substantial effects on a diversity of labour supply indicators (participation, labour earnings, annual weeks and hours worked).[8] Baker et al. (2008), using the first two waves (1994–1995 and 1996–1997) and the last two waves (2000–2001 and 2002–2003) of the NLSCY available at the time, analysed the impact of the childcare policy on formal childcare use and maternal work in two-parent families. Restricting their attention to preschool children aged 0–4 years old, they also showed that the policy had substantial positive effects on mother's employment and nonparental childcare use.

Lefebvre et al. (2009), using data from the SLID (1996–2004), evaluated the potential long-term labour supply effects of Québec's universal childcare policy. They found that the program had substantial dynamic labour supply effects on mothers in Québec, especially for mothers who had a high probability of using low-fee daycare from the child's birth to the fifth birthday. For example, their results show that the policy increased annual hours worked in 2004 by 217 hours for mothers in Québec with at least one child aged 6–11 years old (and no child younger than 6 years, therefore not 'directly' affected by the policy).[9] Their results suggest that the effects were persistent over the life of a child. Kottenlenberg and Lehrer (2013), using the same differences-in-differences (DD) approach as Baker et al. (2008) while also adding waves 6 and 7 of the NLSCY, obtained a larger average effect over the postpolicy period (11 versus 7 percentage points) for the participation rate of mothers with at least one child aged 0–4 years old in two-parent families.

Surprisingly, a number of studies using childcare reforms leading to the expansion of publicly provided childcare in other countries found modest to null effects on maternal employment (e.g., Lundin et al., 2008; Fitzpatrick, 2010; Havnes and Mogstad, 2011; Cascio and Schanzenbach, 2013; Bettendorf et al., 2015). In general, these reforms are not as comprehensive as the Québec childcare reform, which may explain the smaller effects. First, the Norwegian reform studied by Havnes and Mogstad (2011) offered a limited number of spaces, leading to excess demand; it did not cover children younger than 3 years old, and the price remained relatively high compared to the low-fee setting we study. Second, in Lundin et al. (2008) highly subsidised childcare already existed in Sweden for children aged 1–5 years old prior to the reform they studied. As such, Lundin et al. estimated the impact of a price reduction and not the deployment of a low-fee network. Third, Fitzpatrick (2010) and Cascio and Schanzenbach (2013) analysed the impact of prekindergarten programs in America. Both found small to null effects on maternal labour force. While some elements of the American reforms are comparable to the Québec reform, the scope of the reform is narrower as it concerns only children aged 4 years old and the number of hours in care cannot exceed 6 hours per day, compared to 12 hours per day in the Québec context. Finally, Bettendorf et al. (2015) estimated the labour supply effect of both reducing childcare fees and increasing earner income tax credit in the Dutch context. They found small positive impacts on maternal labour supply (participation and hours).

Results of comprehensive childcare reforms are more in line with those of the Québec childcare reform. First, Bauernschuster and Schlotter (2013) studied the German reform that legally entitled children aged 3–6 years old to a space in kindergarten. They found that the reform led to a 14.1 percent increase in maternal employment and a 23.2 percent increase in working hours. The authors attributed the substantial employment effects to the inexistence of private childcare arrangements (absence of crowding-out) prior to the reform and the guaranteed availability of public childcare for all 3–6-year-olds. Second, Hardoy and Schone (2013) studied the more recent public initiatives in Norway that reduced price variations among municipalities and increased childcare coverage for the 1–5-year-olds. They documented a rise in the employment rate of mothers with small children (by 4 percentage points approximately), and positive but small and statistically insignificant results for number of days worked conditional on employment. Third, Nollenberger and Rodrguez-Planas (2014) studied the Spanish reform that extended formal childcare to the 3-year-olds. They found that the average employment of mothers of 3-year-old children increased by 9.6 percent following the reform. The authors also documented that for every 10 children newly enrolled in public daycare, 2 mothers join the labour market. They attributed this somewhat modest effect to the sluggish economic growth in Spain at the time of the reform. Finally, Brilli et al. (2013) estimated the impact of childcare availability (as opposed to a childcare reform) on maternal employment in Italy. They found that a 1 percentage change in public childcare coverage[10] increases maternal employment by 1.3 percentage points and children's language test scores by 0.85 percent of a standard deviation.

In sum, the impact of the Québec reform on mothers' labour force participation is comparable to those of similar reforms in other countries: Norway, 2006; Spain, 1990s; and Germany, 1996. Compared to reforms leading to smaller impacts, these reforms are comprehensive. Typically, they provide more regulated spaces at a reduced fee. They also provide extended services for at least 10 hours per day, they often cover young children (aged 3 years old and younger) and take place in a context where mothers' labour force participation is not extremely high to start with. Other reforms discussed above generally have no more than two of these characteristics, which likely explains the smaller labour supply effects.

3.2. On ECEC and child development

In addition to allowing parents to join the labour force through childcare providers' taking care of their children, subsidised childcare providers are generally endowed with a clear educational mission. Improving the cognitive skills of young children has been suggested as a possible strategy for equalizing opportunities across socioeconomic status – levelling the playing field – and improving children's long-run outcomes. Almond and Currie (2011) argued that formal childcare, as opposed to parental care or informal care, may help reduce the importance of family background on child development and thereby favours equal opportunity. There are several observational studies on the effects of maternal employment or ECEC on child development outcomes (cognitive, behavioural, socioemotional, and health related). We briefly summarise the main evidence and refer

[8] In 2002, the effects on participation, earnings, annual hours and weeks worked of the childcare policy were respectively between 8.1 and 12 percentage points, \$5000 to \$6000 (2001 dollars), 231 to 270 annual hours at work, and 5 to 6 annual weeks of work.

[9] Lefebvre and Merrigan (2008) found the impact of the policy on all mothers with at least one child aged 1–5 years old to be 231 hours in 2002.

[10] Childcare coverage is equal to the ratio of the number of public childcare spaces and the population of children aged 0–2 years old.

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the reader to Almond and Currie (2011) and our Appendix A for more details.

First, on children aged 0–2 years old, a growing body of empirical results indicates that maternal employment and time spent in childcare during the first year of life may have adverse effects on a child's developmental outcomes (such as verbal, reading and math scores; and indices of behavioural problems) observed at later ages (Rhum, 2004; Waldfogel et al., 2002; Brooks-Gunn et al., 2002; Hill et al., 2002, 2005; Gregg et al., 2005).

Second, for children aged 3–5 years old, several studies on the effect of preschool programs found significant positive effects on cognitive outcomes (letter-word identification, spelling and applied problems) and measures of school readiness (Gormley and Gayer, 2005; Gormley et al., 2005; Magnuson et al., 2004, 2007), and especially so for disadvantaged children (e.g., Doyle et al., 2009; Burger, 2010; Sammons et al., 2002, 2003). Felfe et al. (2012), cited above, found that the 1991 Spanish universal childcare reform for 3-year-old children had sizeable positive effects on the PISA reading and math test scores when these children were 15 years old. Cascio and Schanzenbach (2013), also cited above, found that the United States prekindergarten programs (for age 4) have increased preschool enrollment rates of children from both low- and high-income families. They also found that low-income children's test performances increased as late as grade 8, and that the program availability shifted high-income children from private to public preschools, with no impact on overall enrollment and test scores for these children. Finally, the existing American studies on full-day versus half-day kindergarten suggested that the impact of full-day kindergarten on academic and social outcomes is mainly positive (Lee et al., 2006; DeCicca, 2007; Zvoch et al., 2008).

Studies with a compelling structural approach that estimate the impact of childcare and hours of work on child development are not frequent, however one that stands out is Bernal (2008). She found negative effects of childcare and hours of work on cognitive development. We compare our results with hers in Section 7 of this paper.

In sum, it appears that childcare generally contributes positively to the development of children beyond age 2 years old, and especially so for children of less favoured family environments. These results are in stark contrast with current evidence of the Québec childcare reform. Baker et al. (2008), using a DD approach, highlighted the harmful effects of the Québec daycare policy on both parental behaviour and health as well as parent-reported behavioural scores and health indicators for children aged 0–4 years old. The authors suggest that these negative effects may be temporary and associated with the rapid expansion of the childcare system. Kottenlenberg and Lehrer (2013) extended their analysis by using later waves of the NLSCY. They also found negative impacts on children's behavioural and health measures, but nonsignificant results for paternal and family outcomes. Several of the outcomes studied were reported by the parent – in most cases the mother. One may wonder whether the impact of the policy on the mother's participation in the labour force may indirectly affect their perception of problematic behaviour by the child.

This paper contributes to the literature by further documenting, using test scores, the impacts of the Québec daycare reform on children's cognitive development at ages 4–5 years old. For the first time, the impact on 5-year-olds is estimated and all of the 8 available waves of the NLSCY are used. Children observed in later waves were eligible for low-fee daycare from birth, and therefore were treated more intensely than children observed in prior studies.

4. Econometric modeling

To estimate the effects of the daycare reform, we rely on a nonexperimental evaluation framework based on multiple pre- and posttreatment periods. We use a DD model where the treatment group includes Québec children before and after the reform, and the control group includes children of the same age in the RofC observed for the same period . This approach is now well established for evaluating natural experiments (Blundell and Costa Dias, 2009), but has raised a number of concerns (e.g., Bertrand et al., 2004). Concerns related to our application are addressed below.

We use data from all 8 waves of the NLSCY.[11] Waves 1 (1994–1995) and 2 (1996–1997) are considered pretreatment, while waves 3 (1998–1999) to 8 (2008–2009) are considered posttreatment. The program was originally implemented in late 1997. Some children surveyed in wave 2 were interviewed in 1997, but all were interviewed before the implementation of the program. In wave 3, only children aged 3–4 years old were eligible for low-fee daycare, and the number of available spaces was fairly constrained. Starting in wave 4 (2000–2001), all children residing in Québec were eligible. It is also around that time, that the 'constraint' on the number of low-fee spaces was loosened by the authorities. In the spirit of Francesconi and van der Klaauw (2007), we write down a model that accounts for the relatively long implementation period. The empirical model is as follows:

$$Y_{ij} = \alpha + \theta Q_{ij} + \sum_{j=3}^{8} \beta_j W_j Q_{ij} + \sum_{j=1}^{8} \gamma_j D_j + \Phi X_{ij} + \varepsilon_{ij}. \tag{1}$$

where Y_{ij} represents the outcome of child i (or the child's mother or father) in wave j. We study the following outcomes: weekly hours in childcare, mother's and father's labour force participation and weeks worked, and the cognitive test scores of children at ages 4 and 5 years old. The term Q_{ij} takes the value of 1 if the child i lives in Québec in wave j, and otherwise takes the value 0. A set of eight D_j cycle dummy variables capture aggregate effects. To account for the progressive implementation of the policy, a set of dummies W_j for each of the postreform waves $j = 3,4,5,6,7,8$ interacted with Q_{ij} is included. This allows the effect of the reform to be different over time. Between wave 3 (1998–1999) and wave 8 (2008–2009), more than 110,000 new low-fee spaces were created (an average of 14,000 per year). The term X_{ij} is a vector of socioeconomic control variables and Φ is a vector of parameters. This allows us to control for socioeconomic changes in group composition. Finally, ε_{ij} is an i.i.d. error term. Assuming the effects of the policy are heterogeneous, we estimate the Average Treatment on the Treated effect (ATT).

To account for unobserved transitory shocks at the group level, we also implement a two-step procedure to adjust the standard errors (Donald and Lang, 2007). Effectively, we first regress the outcome variables on a set of 16 dummies, one for each region-wave interaction, and X_{ij}, while taking into account the population weights. Then, we regress the estimated coefficients of the 16 interaction terms on a constant, 7 cycle dummies, a Québec dummy and the 6 interaction terms $W_j Q_{ij}$. Each observation is weighted by the inverse of the variance of the estimated interaction term. According to Donald and Lang, inferences should be conducted using a student's t-distribution with 1 degree of freedom (t_1-distribution), but since our group sizes are large, we follow Wooldridge (2006) and use the standard normal distribution for inference.

The RofC control group accounts for other Canadian reforms that have an impact on parental labour supply and child development such as the introduction of the federal National Child Benefit Supplement (NCBS) in 1998 that replaced the Working Income Supplement for low-income families. The NCBS effectively increased the benefits per child paid to low-earner families.[12]

Although Canada-wide reforms can be accounted for using our empirical approach, Québec specific reforms threaten the validity of

[11] The data set is discussed further in the next section.

[12] Milligan and Stabile (2007) found that this reform had a small significant positive effect on the labour supply of low-income families in Canada for 5 provinces (excluding Québec) and their overall income. Impacts were stronger for single-parent families, but extremely modest for two-parent families.

our estimates. Two reforms were implemented during the period we observe. First, in July 1997, universal nontaxable family allowances were replaced by a tax benefit contingent on family income as well as family status. Second, in January 2005, Québec implemented a fiscal reform for families with children that included a new working income supplement to low-income households (mostly favouring single-parent families working near the minimum wage). This reform increased the disposable income of low-income families. We discuss further, in Sections 6 and 7, the implication of these reforms with respect to the daycare reform.

For the estimation, we use different subsamples of the NLSCY waves 1–8 data set. We first selected all children aged 1–5 years old. Children living in foster families are excluded as well as those (very few) with a mother observed with missing values for the socioeconomic control variables. This constitutes our main sample. From this sample, a number of subsamples are created. The main sample is divided by age of the child and by maternal education (those with some postsecondary education or less, and those with a postsecondary diploma or higher). We specify in the tables of results which subsample is used for each estimation and provide bootstrapped standard errors (including for the two-step procedure) using the 1000 bootstrap weights that allows treatment of the data as repeated cross-sections provided by Statistics Canada to account for the complex survey design of the NLSCY. It is always possible that prepolicy trends differed in Québec and the RofC. For the case of labour supply, Lefebvre and Merrigan (2008) showed that the introduction of trends did not affect the estimated effects when the policy was still young. The data used in that paper included 5 prepolicy years of data, the NLSCY has only 2 prepolicy years of data making regional specific pretrend analysis less credible.

The following sections present the empirical analysis. First, we present the data. Second, we discuss the impact of the reform on hours of childcare, and present the estimated impacts on labour force participation and weeks worked. Finally, we display the results on cognitive test scores.[13]

5. Data set

For this study, we use the NLSCY, a biennial survey conducted by Statistics Canada, started during 1994–1995 (wave 1) and ending during 2008–2009 (wave 8).[14] The survey asks Canadian parents a wide range of questions on the use of childcare and their labour force participation, [15] which allows us to trace their evolution in Québec and the RofC using comparable data. The unit of analysis for the NLSCY is the child or youth. The NLSCY not only provides information on representative provincial samples of all children, but in contrast with administrative data, it also provides information on both regulated and nonregulated (more informal) types of care.

Table 1 presents the descriptive statistics for children aged 1–5 years old by region (Québec and RofC) and by wave (1994–2008). Generally speaking, the observed characteristics are fairly comparable across region and years, but a few differences and trends are worth mentioning. While mothers' education was comparable in 1996 in Québec and the RofC, mothers in Québec became more educated over time with 68.6% receiving a postsecondary diploma or more in 2008 compared to 59.6% in the RofC. However, except for the last two years of data, the trends in education were similar.[16] Over the entire observation period, there are relatively more immigrant mothers in the RofC by approximately 9 percentage points, and more mothers giving birth at 35 years of age or older by approximately 4 percentage points. Family composition is comparable and relatively stable over time. A large fraction of children in Québec and the RofC live in a large urban area (more than 500,000 inhabitants). Finally, family total gross income ($2002) is higher in the RofC, but the gap is fairly stable over time.

We show above that the care of young children in Québec has changed in important ways. These changes become even more dramatic when the evolution of the use of childcare in Québec is compared with that of the RofC using the NLSCY data set. Fig. 3 summarises, for Québec and the RofC, the four principal care arrangements used by parents for children aged 1–4 years old, from 1994–2008. The vertical line indicates the third wave of the survey (1998), which corresponds to the second year of policy implementation. Prior to the daycare reform, the care arrangements in Québec and the RofC were fairly comparable. Most children were in parental care (around 55%). Some were in out-of-home care [17] (around 25%), and a comparable fraction (around 10%) were either in centre-based care or were cared for at home but not by their parents. Home care is slightly higher outside Québec, while centre-based care is slightly higher in Québec. Starting in 1998, the use of centre-based care increased rapidly in Québec while parental care (and to a lesser extent out-of-home care) decreased rapidly. In contrast, in the RofC, no such dramatic changes are noticeable. Parental care remained fairly stable at approximately 50% since 1998. After parental care, family-based care outside the child's own home remains the most widely used mode of care in the RofC for most of the period. To summarise, the figure shows an important shift in daycare use occurring in Québec after the introduction of the daycare policy in 1997. No comparable shift is observed in the RofC.

The change in care arrangements is also combined with a change in the intensity of care. Fig. 4 presents the mean hours (conditional and nonconditional on the use of childcare) per week that children spent in their primary care arrangement in Québec and the RofC. Nonconditionally, prior to the childcare reform, children aged 1–4 years old in both regions spent on average 11–12 hours per week in childcare. Starting in 1998, the average hours children spent in daycare in Québec started to increase and continued to do so for the entire observation period. By 2008, children in Québec spent on average 23 hours per week in daycare, while children in the RofC spent 12–13 hours per week in daycare. Conditional on attending daycare, we find that children in Québec prior to the reform were already in care for longer hours: 29–30 hours compared to 27 hours in the RofC. This difference becomes even more striking as the reform's implementation progressed. During 2006–2008, children in Québec attending daycare spent on average 33 hours per week in daycare compared to 28 hours per week in the RofC. This figure shows not only that more children started to attend daycare in Québec following the reform, but that the intensity of care for those attending daycare also increased.

To summarise, the figures presented so far show important shifts in daycare use, modes, and intensity occurring in Québec after the introduction of the childcare policy in 1997, but not in the RofC. They are consistent with the DD assumption that no specific child care policies were introduced in the RofC for the years of sample. They also show that the change in hours in care follow very closely the increase in subsidised spaces starting in 1998.

6. Results for hours of care and parental labour force participation

6.1. Children aged 1–4 years old

We start by providing evidence on the effects of the program on hours of care and parental labour supply for children ages 1–4 years

[13] The fiscal implications of the policy are discussed in the last section of our Appendix A.

[14] From here on, we refer to " years" instead of " waves," and use only the first year of the two-year sample period. As such, we refer to 1994 for wave 1, 1996 for wave 2, and so on.

[15] A parent is participating in the labour force if the parent answers yes to the question 'Do you currently work?' Results are robust to the use of another question 'Did you work in the past 12 months?'

[16] In Québec, postsecondary institutions include not only universities and trade schools, but also the CEGEP (Collège d'Enseignement Géné ral et Professionnel). Graduating from CEGEP takes only two years, and it is a necessary condition for admission to a university. In the RofC, students enter university directly after high school.

[17] Out-of-home care includes family-based care (regulated or not) and care by relatives outside of the child's home.

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Table 1
Mean characteristics of all mothers and families with at least one child aged 1–5 by region and wave.

Year	1994	1996	1998	2000	2002	2004	2006	2008
Québec								
Child is a girl	51.8	50.5	51.8	51.4	51.4	51.7	51.7	51.8
Mother								
Less than high school	18.5	15.2	16.0	15.4	14.7	18.9	10.0	7.9
High school diploma	16.2	15.5	12.4	17.7	19.0	16.4	10.8	12.3
Some postsecondary	25.9	23.4	24.5	19.7	19.3	11.8	15.8	11.9
Postsecondary diploma	39.4	46.0	47.1	47.2	47.0	52.9	63.4	67.9
Age 14–24 at birth	20.6	20.8	21.0	25.7	25.5	23.9	18.4	16.0
Age 25–29 at birth	40.6	37.4	34.9	34.5	34.1	36.6	38.5	38.2
Age 30–34 at birth	28.9	31.2	32.5	27.8	27.6	26.5	29.0	32.9
Age 35 or more at birth	9.9	10.7	11.5	12.0	12.9	13.0	14.1	12.8
Born in Canada	91.6	91.8	89.2	88.2	86.4	82.8	82.2	81.8
Family								
Single parent	12.9	14.1	14.0	15.0	11.9	13.8	12.5	11.2
Two-parent	87.1	85.9	86.0	85.0	88.1	86.2	87.4	88.8
Older siblings	49.3	51.5	56.8	55.4	52.0	53.6	53.6	50.9
Younger siblings	31.3	27.4	25.9	27.0	28.2	25.6	25.3	30.1
Presence of other children	0.4	1.1	1.6	0.5	0.6	0.2	0.5	0.2
Size of area of residence >499 K	53.9	59.1	60.5	58.8	55	57	62.2	62.3
Size of area of residence 100–499 K	11.8	6.1	6.2	6.2	5.8	4.7	7.1	7.0
Size of area of residence 30–99.999 K	8.4	9.1	9.6	9.1	8.3	7.2	10.5	9.6
Size of area of residence < 30 K	8.2	11.2	10.3	9.9	17.5	17.7	6.6	6.9
Size of area of residence: rural	17.7	14.4	13.4	16.0	13.4	13.4	13.6	14.2
Unemployment rate	12.9	13.2	9.7	9.5	9.5	10.4	8.8	9.3
N	1 797	1 666	3 472	2 460	1 900	1 309	1 298	1 402
Rest of Canada								
Child is a girl	50.9	51.6	51.5	51.1	51.1	51.3	51.2	51
Mother								
Less than high school	14.4	11.5	10.8	12.4	10.5	10.0	8.5	9.1
High school diploma	18.3	18.2	16.8	18.9	23.1	22.0	17.8	17.8
Some postsecondary	28.9	28.9	27.0	21.7	13.6	12.3	12.3	13.8
Postsecondary degree	38.4	41.3	45.4	46.9	52.8	55.7	61.4	59.4
14–24 years at birth	19.3	18.4	17.7	21.0	19.0	19.3	18.6	18.8
25–29 years at birth	36.4	35.6	31.6	32.1	32.1	28.9	28.7	29.3
30–34 years at birth	30.7	32.3	34.4	31.8	32.1	33.2	33.8	33.1
35 years or more at birth	13.6	13.6	16.3	15.1	16.8	18.6	19.0	18.8
Born in Canada	79.5	78.8	79.7	78.9	74.5	73.3	72.3	72.4
Family								
Single parent	15.7	14.8	13.9	14.1	12.7	12.5	13.6	13.5
Two-parent	84.2	84.8	86	85.7	87.1	87.4	86.3	86.3
Older siblings	57.0	56.0	58.4	55.8	56.9	56.0	55.2	55.6
Younger siblings	29.9	29.7	29.6	28.6	26.9	26.2	26.6	27.4
Presence of other children	1.7	1.4	1.7	0.7	0.9	1.6	1.8	1.8
Size of area of residence >499 K	41.5	42.7	44.2	46.4	42.6	42	49.2	48.1
Size of area of residence 100–499 K	20.8	22.1	21.6	21.5	15.0	14.7	20.9	21.2
Size of area of residence 30–99.999 K	7.4	8.3	8.8	8.7	8.7	8.6	9.2	9.5
Size of area of residence < 30 K	12.9	14.8	13.7	12.9	23.6	24.5	9.0	9.4
Size of area of residence: rural	17.5	12.1	11.7	10.5	10.2	10.1	11.6	11.9
Unemployment rate	9.4	8.6	7.3	6.2	6.8	6.1	5.4	7.7
N	7 851	7 114	13 456	12 511	9 334	7 432	8 477	9 019

Note: Shows the weighted summary statistics for children aged 1–5 by year. All statistics are percentages except household income.

old. Relative to former studies, here, we simply extend the observation period to assess whether the leading effects identified by previous studies are persistent. Compared to previous work that estimated the impact of the policy on participation in childcare, we add some results on the occurrence and intensity of care using hours in care both conditionally and unconditionally on attending any form of nonparental care.

Table 2 shows the estimated effects of the policy. Our evaluation strategy relies on Eq. (1). Four outcomes Y_{ij} are presented: the number of hours per week spent in the primary mode of care for all children (columns 1 and 2) and for children attending daycare (columns 3 and 4), the current labour force participation of the mother at the time of the survey (columns 5 and 6), and the number of weeks worked by the mother in the reference year (columns 7 and 8). Estimates presented in columns 2, 4, 6 and 8 are computed with the two-step procedure detailed in the previous section. The socioeconomic controls X_{ij} include the following: (1) the sex of the child; (2) child age dummies; (3) the age group of the mother at child's birth (25–29, 30–34, 35 or older; with 14–24 as the omitted group); (4) the family type (single-parent, with two-parent family as the omitted group); (5) a dummy for whether the mother was born in Canada; (6) the mother's highest level of education (less than a high school diploma, high school diploma, some post-secondary education, trade or college diploma, with a university diploma or more as the omitted group); (7) the presence of siblings (older, younger or same age, and the omitted group is children with no sibling), and other children; (8) the size of the community of residence (5 groups ranging from rural to 500,000 inhabitants; the omitted group is those with more than 500,000 inhabitants); and (9) the provincial unemployment rate. These characteristics are presented in Table 1 for children aged 0–5 for both regions.

For unconditional hours of care (including zero hours), the estimated effects increased substantially from 2 hours in 1998 to 11 hours in 2008 (column 1). The effects are significant and remain so once we account for unobserved aggregate transitory shocks (column 2). In general, the estimation shows that the effects increased over time, reflecting the

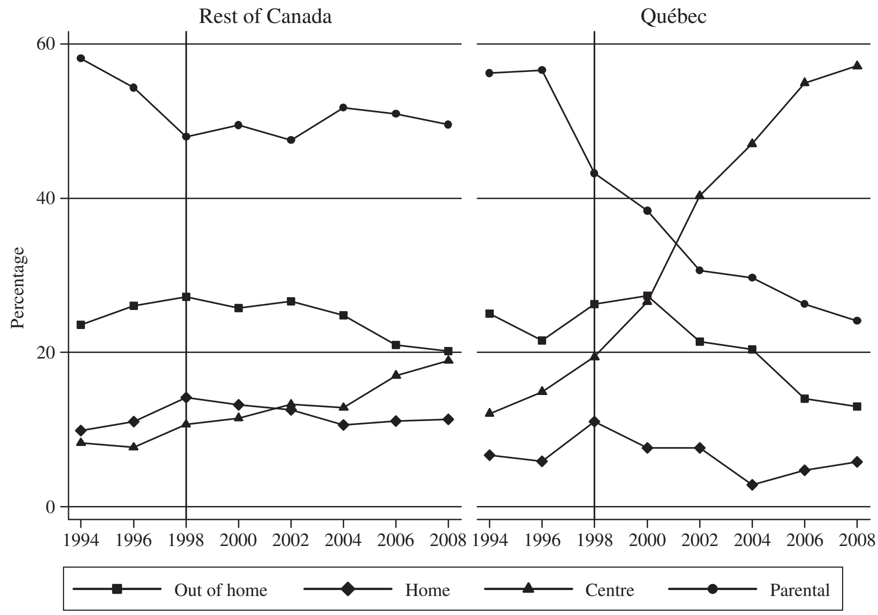

Fig. 3. Primary care arrangement of children aged 1–4 by year in percentages. Note: Shows the evolution of the percentage of the primary modes of care in the Rest of Canada (left panel) and Québec (right panel) by wave. The sample includes NLSCY cross-sectional children aged 1–4.

addition of new low-fee spaces. These results imply that while all children postreform spent more time in daycare, children observed in the later years of the reform were even more exposed to daycare. Clearly, these effects show a dramatic and significant change in the lives of children in Québec compared to children in the RofC.

The unconditional number of hours certainly increased due to the higher daycare participation rate postreform (see Fig. 3), but also possibly due to the longer hours of services offered by the subsidised providers. To measure whether children in care were there for longer hours postreform, we estimate the impact on hours of care conditional on attending daycare (i.e., we now exclude children with hours of care from our sample). We find that conditional on attending daycare children spent more time in care postreform. In the first year postreform (1998), we find that children spent an additional 2.1 hours per week in daycare. By 2008, this effect had reached 3.8 hours more per week, with a peak at 5.2 hours in 2004. Not only did the reform induce childcare participation, but it also increased the intensity of care for those participating.

Table 2 shows the estimated effects on maternal labour force participation (columns 5 and 6) and weeks worked (columns 7 and 8). As of 2000, the estimates on labour force participation are large and significant. In 2000, the estimate suggests an increase of 5 percentage points.

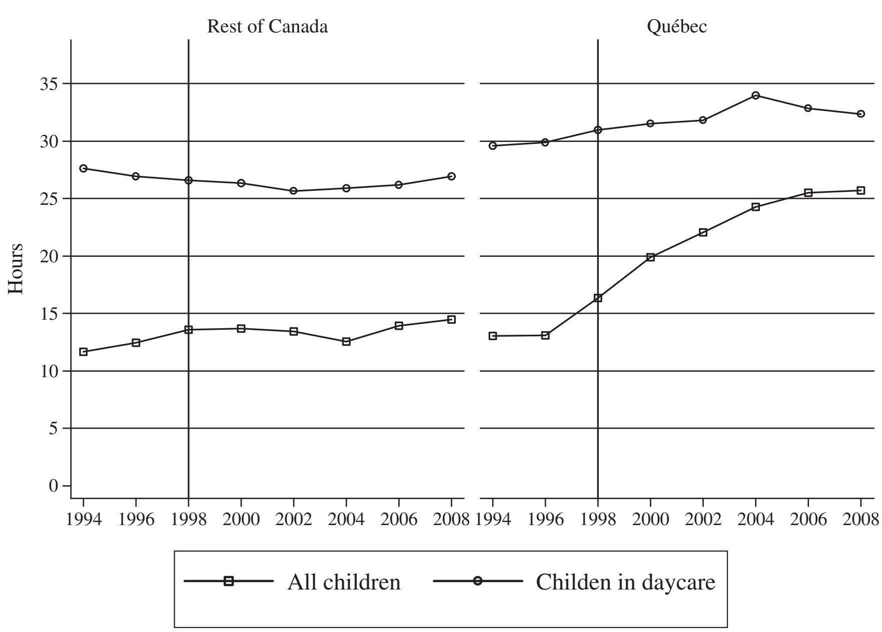

Fig. 4. Mean hours per week spent in the primary care arrangement for children aged 1 to. Note: Shows the evolution of the mean number of hours per week spent in the primary mode of care in the Rest of Canada (left panel) and Québec (right panel) non conditionally (hollow square) and conditionally on attending childcare (hollow circle). The sample includes NLSCY cross-sectional children aged 1–4.

Table 2
Estimated effects of the policy on weekly hours spent in the primary mode of care: children aged 1–4.

Specification	Hours of care (1)	Hours of care (2)	Conditional hours of care (intensity) (1)	Conditional hours of care (intensity) (2)	Mothers' labour force participation (1)	Mothers' labour force participation (2)	Mothers' weeks worked (1)	Mothers' weeks worked (2)
β_3 (1998–99)	2.31***	2.28***	2.09**	2.09**	−0.00	−0.00	−0.35	−0.35
	(0.85)	(0.85)	(0.99)	(1.00)	(0.02)	(0.02)	(1.07)	(1.06)
β_4 (2000–01)	5.63***	5.63***	2.98***	2.98***	0.05**	0.05**	1.58	1.61
	(0.83)	(0.83)	(0.91)	(0.91)	(0.02)	(0.02)	(1.12)	(1.12)
β_5 (2002–03)	8.68***	8.69***	4.47***	4.47***	0.08***	0.08***	4.02***	4.05***
	(0.90)	(0.90)	(0.93)	(0.94)	(0.02)	(0.02)	(1.12)	(1.12)
β_6 (2004–05)	10.92***	10.92***	5.24***	5.24***	0.13***	0.13***	6.18***	6.21***
	(1.00)	(1.00)	(0.95)	(0.95)	(0.02)	(0.02)	(1.23)	(1.22)
β_7 (2006–07)	11.34***	11.35***	4.44***	4.45***	0.10***	0.10***	5.04***	5.07***
	(0.89)	(0.89)	(0.88)	(0.88)	(0.02)	(0.02)	(1.28)	(1.28)
β_8 (2008–09)	10.74***	10.75***	3.82***	3.82***	0.12***	0.12***	6.21***	6.24***
	(0.92)	(0.92)	(0.95)	(0.96)	(0.02)	(0.02)	(1.25)	(1.24)
N	65,248	16	34,511	16	65,880	16	65,806	16

Note: We use the sample of all children aged 1–4 years old. The dependent variables are the number of hours in care (columns 1–4), a dummy for labour force participation at the time of the interview (columns 5–6) and the number of weeks worked in the reference year (columns 7–8). All specifications include cycle dummies and control for the sex of the child, the age of the child in years, the age group of the mother at child birth (25–29, 30–34, 35 or more with 14–24 the omitted group), the family type (single parent, with two-parent the omitted group), a dummy for whether the mother was born in Canada or not, the mother's highest level of education (less than a high school diploma, high school diploma, some postsecondary education, with postsecondary diploma the omitted group), the presence and number of older or younger siblings or the presence of a child of the same age, the size of the community (five groups from rural to 500,000 or more the omitted group) and the provincial unemployment rate. Specification (1) relies on Eq. (1), while specification (2) relies on the two-step procedure to account for unobserved aggregate transitory shocks. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is $p < 0.01$, ** is $p < 0.05$, and * is $p < 0.1$.

In line with the addition of new low-fee spaces, this effect slightly increased up to 13 percentage points in 2004 and remained close to this level until 2008. On weeks worked, we also observe a sustained positive policy effect from 2000–2008. These findings imply that the positive impact on labour force participation dominates the income effect of the low-fee network. The standard labour supply model predicts that, due to the income effect, mothers active in the labour market could decrease their number of weeks worked following the implementation of the low-fee program. However, because the low-fee program also resulted in a large increase in the labour force participation of mothers, a large fraction of mothers moved from working zero weeks to working a positive number of weeks. Since the labour force participation effect is so large, it dominates the income effect, which is why we observe positive effects on weeks worked.

In sum, the effects are generally positive and significant on both conditional and unconditional hours in care. In contrast with previous work that focused exclusively on daycare participation, this further shows that not only did participation increase, but so did the number of hours spent in care conditional on attending daycare. In line with previous work, we also find that the reform has a positive impact on maternal labour force participation and weeks worked, but we also show that these effects generally increased in the early years and then persisted until 2008. The flattening of the effects as of 2004 is paralleled in part by a smaller progression in the number of spaces between 2004 and 2008 (see Fig. 1). When the labour force participation rate becomes rather high, it is more difficult to get more mothers in the labour market, either because they have strong preferences for staying home or they have low skills and cannot be hired.

6.2. Hours of care by age

We now present the effects of the reform by age of the child and maternal education. We start with hours of care, and then present the results related to the labour supply of mothers. We analyse 5-year-olds eligible for kindergarten separately from children of that age who are not age-eligible for kindergarten. This allows us to estimate the effect of the full-day kindergarten policy. Table 3 shows the estimated effects of the policy on weekly hours in childcare by age (1–5 years old). We start with the effect on hours of care not conditional on attending daycare (top panel), and then present the effect on hours of care conditional on attending daycare (bottom panel).

For ages 1–4 years old, we find that the number of hours spent in care increases during the period across all age groups. It is not significant in 1998 for children aged 1–3 years old. At this time, children aged 0–2 years old were not yet eligible and 3-year-olds had only been eligible for a few months. For 4-year-olds, eligible since September 1997, we estimate large positive effects as early as 1998. As of 2000, when all children were eligible, we observe large positive effects across all age groups, and these generally increase over time and also increase with age. By the end of our observation period, children aged 1–4 years old spend an additional 6.8 and 15.1 hours, respectively, in care.

For children age 5 years old who are not yet age-eligible for kindergarten, we find that hours in care increased significantly as a result of the daycare reform. As of 2000, these children spend 6.1 hours more in daycare per week, and by 2008 they spend 16.8 hours more. This effect is very large. The larger effects are found in 2006 and 2008 (16.2 and 16.8 hours respectively). These children were the first to be eligible for low-fee daycare since birth. For 5-year-olds who were age-eligible for kindergarten, the effects are negative and significant as early as 1998. As these children switched from half-day kindergarten to full-day kindergarten (as of 1998), they only attended daycare before and after school (if they did at all). This effect is more or less constant and ranges from −3.6 to −5.2 hours per week between 1998 and 2008.

Again, to measure the impact on the intensity of care for children attending daycare, we estimate the effects of the reform on the sample of children with a positive number of hours in care (i.e., hours > 0, bottom panel of Table 3).[18] For children aged 1–2 years old, the intensity of care increased between 2000 and 2004, but eventually returned to prereform levels in 2006 and 2008. This reflects a change in the behavior of parents of young children. We summarise two possible changes in behaviour: (1) an increase in the participation of parents using childcare services on a part-time basis; and (2) an overall reduction in hours of care for young children. Observing the participation pattern for these age groups,[19] we find that participation did not increase between 2004 and 2006 for 1-year-olds. This suggests that our second explanation likely explains the changing pattern for this age group. For 2-year-olds, participation in care did increase between 2004 and 2006.

[18] The estimate should be downward biased because the post-treatment sample should on average contain more mothers with a lower attachment to the labour market.
[19] These results can be obtained from the authors on request.

Table 3
Estimated effects of the policy on weekly hours spent in the primary mode of care by age of the child.

Age of children	1	2	3	4	5 not eligible for K-5	5 eligible for K-5
Hours of care						
β_3 (1998–99)	2.44**	2.73	0.95	3.13*	1.79	−5.21***
	(1.13)	(1.87)	(1.69)	(1.83)	(2.29)	(1.17)
β_4 (2000–01)	4.54***	8.11***	5.30***	4.85**	6.11**	−4.33***
	(1.50)	(1.56)	(1.39)	(1.90)	(2.58)	(1.45)
β_5 (2002–03)	8.13***	10.29***	8.17***	7.81***	11.89***	−4.39***
	(1.82)	(1.85)	(1.69)	(1.64)	(2.38)	(1.20)
β_6 (2004–05)	11.08***	12.05***	9.22***	11.50***	6.65*	−4.67***
	(1.63)	(1.94)	(1.94)	(2.24)	(3.86)	(1.37)
β_7 (2006–07)	7.72***	10.32***	12.60***	14.41***	16.25***	−3.87***
	(1.69)	(1.85)	(1.64)	(1.81)	(3.67)	(1.45)
β_8 (2008–09)	6.84***	10.00***	11.66***	15.07***	16.76***	−3.59***
	(1.57)	(2.14)	(1.69)	(1.81)	(3.18)	(1.38)
N	19,363	13,978	18,105	13,802	6594	16,424
Conditional hours of care (intensity)						
β_3 (1998–99)	−0.75	−0.05	2.46	7.41***	2.93	−8.81***
	(1.34)	(2.09)	(1.84)	(1.85)	(2.83)	(1.62)
β_4 (2000–01)	2.86**	3.23*	3.03***	3.63*	2.54	−9.33***
	(1.43)	(1.74)	(1.39)	(1.99)	(3.60)	(1.80)
β_5 (2002–03)	2.07	6.11***	4.79***	5.01***	7.39**	−9.26***
	(1.70)	(1.84)	(1.68)	(1.65)	(3.19)	(1.67)
β_6 (2004–05)	4.21***	6.00***	3.85**	8.13***	12.11**	−9.66***
	(1.55)	(1.88)	(1.92)	(1.99)	(3.80)	(1.81)
β_7 (2006–07)	0.34	1.31	5.60***	10.28***	7.99**	−11.52***
	(1.55)	(1.87)	(1.57)	(1.72)	(3.89)	(1.92)
β_8 (2008–09)	0.09	2.07	5.74***	8.07***	13.43***	−10.31***
	(1.59)	(2.09)	(1.66)	(2.00)	(3.37)	(1.88)
Observations	9605	7398	10,065	7443	3252	8005

Note: The dependent variables are the number of hours in care including zero hours of care (top panel) and excluding zero hours of care (bottom panel). The age of children included in the sample is specified at the top of each column, with children aged 5 years old divided in two groups: (1) not age eligible for kindergarten (K-5) and (2) age eligible to K-5. Children from all family types are included. We include the same controls as in Table 2. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is $p < 0.01$, ** is $p < 0.05$, and * is $p < 0.1$.

Table 4
Estimated effects of the policy on mothers' labour force participation and weeks worked by age of the children.

Age of children	1	2	3	4	5 not eligible for K-5	5 eligible for K-5
Mothers' current labour force participation						
β_3 (1998–99)	0.04	0.02	−0.01	−0.04	−0.12*	−0.04
	(0.03)	(0.05)	(0.04)	(0.04)	(0.07)	(0.04)
β_4 (2000–01)	0.08**	0.09**	0.01	0.02	−0.13*	−0.02
	(0.04)	(0.04)	(0.04)	(0.05)	(0.07)	(0.05)
β_5 (2002–03)	0.15***	0.08*	0.05	0.03	0.01	−0.01
	(0.04)	(0.05)	(0.05)	(0.04)	(0.07)	(0.05)
β_6 (2004–05)	0.16***	0.14***	0.14***	0.07	−0.03	0.10*
	(0.04)	(0.05)	(0.05)	(0.05)	(0.09)	(0.06)
β_7 (2006–07)	0.13***	0.11*	0.14***	0.03	0.14	0.12**
	(0.04)	(0.06)	(0.04)	(0.05)	(0.10)	(0.05)
β_8 (2008–09)	0.07	0.19***	0.10**	0.12**	0.01	0.09*
	(0.04)	(0.05)	(0.04)	(0.05)	(0.09)	(0.05)
N	19,650	14,114	18,210	13,906	6668	16,608
Mothers' weeks worked in reference year						
β_3 (1998–99)	1.79	0.52	−0.89	−2.44	−3.30	−2.63
	(1.57)	(2.53)	(2.13)	(2.00)	(3.36)	(2.12)
β_4 (2000–01)	3.11	2.15	−0.06	1.45	−5.27	−2.37
	(2.03)	(1.95)	(1.80)	(2.63)	(3.69)	(2.52)
β_5 (2002–03)	8.64***	2.69	1.72	3.08	2.17	−1.60
	(2.22)	(2.29)	(2.32)	(2.06)	(3.73)	(2.35)
β_6 (2004–05)	6.45***	5.69**	6.97***	6.21**	2.76	3.39
	(2.16)	(2.36)	(2.35)	(2.46)	(4.81)	(2.81)
β_7 (2006–07)	4.10*	5.50**	6.95***	3.75	8.40*	4.66
	(2.09)	(2.77)	(2.21)	(2.67)	(5.00)	(2.85)
β_8 (2008–09)	5.12**	8.77***	3.91*	7.55***	3.90	3.07
	(2.10)	(2.62)	(2.00)	(2.52)	(4.45)	2.89)
Observations	19,585	14,101	18,202	13,918	6673	16,589

Note: The dependent variables are a dummy for maternal labour force participation (top panel) and the number of weeks worked by the mother in the reference year (bottom panel). Children from all family types are included. We include the same controls as in Table 2. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is $p < 0.01$, ** is $p < 0.05$, and * is $p < 0.1$.

This may imply that new users of childcare services may have been part-time users.

For older children, the effects on conditional hours in care are larger and persistent over time. Children 3 years old spent on average an additional 4–6 hours per week in care postreform, while 4-year-olds spent as much as 10 more hours per week and 5-year-olds (not yet in school) spent 13 more hours. Again, these results suggest that not only did the number of children in care increase, but also the intensity of care for children aged 3–5 years old. For 5-year-olds in school, the number of hours in care (conditional on using nonparental care) decreased substantially as of 1998 (8.8 hours per week). This reflects the implementation of full-day kindergarten as of 1998. Children 5 years old who were in school attended kindergarten all day, and were in care only before- and after-school. Not surprisingly, they spent considerably less time in care: their hours of care decreased by 8.8 hours in 1998 up to 10 hours in 2008.

We estimated this specification with the subsample of children whose mothers have a some postsecondary education or less and the subsample of children whose mothers have a postsecondary diploma or higher.[20] We find that the effects are generally comparable in both groups, but the results by age group for children of less-educated mothers are slightly less stable, probably due to the smaller sample size. For the aggregated group of 1–4-year-olds, the effects are increasing, from small and not significant in 1998–10.9 hours in care for the highly educated and 9.6 for all other mothers in 2008. In sum, regardless of the education of the mother, children from 2004–2008 were more intensely affected by the program than children observed earlier, which is consistent with the increased number of low-fee spaces.

6.3. Mothers' labour force participation

The top panel of Table 4 shows the evolution of the impact of the reform by age group on mothers' labour force participation, while the bottom panel shows the impact on weeks worked. Weeks worked is measured over the past 12 months and includes weeks on paid maternity or parental leave.

For labour force participation, we find that the reform generally has a positive, large and significant effect on the labour force participation of mothers with a child of 1–4 years old. More specifically, we find that for children aged 1–2 years old, results are in line with the policy implementation with a clear jump as of 2000. For children aged 3–4 years old, the effect is larger as of 2000, but only becomes significant as of 2004 (age 3) and 2008 (age 4). This suggests that the policy mainly raised the labour force participation of new mothers, those giving birth after the policy change. Mothers of 3-year-olds and older at the time of the reform had already interrupted their career for a number of years. This may have limited their opportunities to reenter the labour market as their human capital had likely depreciated. The pattern for weeks worked by the mother is consistent with those of labour force participation. For the 1–4-year-olds we generally find positive and

[20] The results are presented in Table A.2 in our Appendix A. Postsecondary diploma includes college, trade and university.

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significant effects. We observe a monotonic increase from 1998–2008 for children aged 2–4 years old. The pattern for 1-year-olds also increased but only until 2004, then the effect slightly decreased and stabilised in cycles 7 and 8. This in line with the previous finding on labour force participation.

The effects on the labour force participation of mothers with a 5-year-old not yet age-eligible for kindergarten are not significant. The number of weeks worked, however, increased over time, following a pattern similar to the number of hours of care for this group. While the number of hours of care was larger for these children, their mothers did not participate more in the labour market. Mothers of 5-year-olds eligible for kindergarten significantly increased their labour force participation as of 2004 (0.10). The pattern on weeks worked is comparable, but not significant. Although their mothers worked more, these children effectively spent less time in daycare. What is particular about the children from the later waves is that the number of childcare spaces available was substantially higher for them since birth. Therefore, the positive (and sometimes significant) effects at age 5 in the later waves could be due to the availability of low-fee spaces since birth. Therefore, simply changing kindergarten policy from a part-time to a full-time system may not be enough to increase the labour supply of mothers with a 5-year-old if it is not combined with a daycare policy for the very young. This is in line with previous findings in Fitzpatrick (2010) on the impact of adding one more year of full-time schooling (in this case prekindergarten).

Table 5 presents the results on labour force participation and weeks worked for mothers with some postsecondary education or less (panel one) and with a postsecondary diploma or more (panel two). Starting with mothers with some postsecondary education or less, we find that the estimated effects on labour force participation are generally not significant by age group, except for 2-year-olds. Once we aggregate mothers of children aged 1–4 years old, we find stronger effects for the last three periods consistent with the increase in subsidised spaces. For highly educated mothers, the evidence is different. The effects are positive, large and significant for children aged 1–4 years old following the pattern described above using 4. When we compare mothers of children aged 1–4 years old (as an aggregate) by education level, we find that the effects are slightly larger and are positive and significant earlier for highly educated mothers. Two reasons may explain this slight difference. First, the earning potential of highly educated mothers is generally higher, such that the incentive to return to work is also generally higher. Second, as mentioned above, low-income households were eligible for a childcare subsidy prior to the reform such that the net cost did not change drastically for them over our observation period. In contrast, prior to the reform highly educated mothers with high potential earnings were not eligible for the childcare subsidy. For them, the implementation of low-fee daycare constituted a strong incentive to return rapidly to the labour market.

The results on weeks worked by education (panels three and four of Table 5) mimic quite closely those found on participation. For mothers with some postsecondary education or less (third panel) and children aged 1–4 years old, the policy also has mainly positive but not significant effects. These effects are generally smaller (and less significant) than for the well educated (fourth panel).

In sum, we find that the effects on hours of care, labour force participation and weeks worked were generally stronger in the later years of the reform, generally positive and increasing in time for children aged 1–4 years old. Older children in the later waves were eligible for low-fee daycare since birth. Because of the implementation of full-day kindergarten, the effects on hours of care are negative and significant in all years for children eligible for kindergarten, and positive on labour force participation in later waves, which suggests that low-fee daycare (as opposed to full-day kindergarten) triggered these effects. Postsecondary educated mothers responded more strongly to the reform early on. In line with the progressive implementation of the reform the evidence presented here suggests that the effect is increasing with the number of subsidised spaces.

6.4. Fathers' labour force participation and weeks worked

There is no evidence, to our knowledge, of the impact of daycare reforms on fathers' labour force participation. Fig. 5 presents descriptive statistics for children aged 1–4 years old from the 8 waves of the NLSCY on the labour force participation of fathers. From 1998–2008, the changes for the two indicators were marginal: labour supply was high at around 93 percentage points. For weeks worked, we observe the same patterns. [21] Table 6 shows the estimated effects on fathers' labour force participation (top panel) and annual weeks worked (bottom panel). For each age group, the effects are rarely significant and relatively small. These results are also robust to the two-step procedure accounting for unobserved transitory shocks. [22] As for weeks worked (second panel, Table 6), we find that over the years, fathers in our sample barely changed their number of weeks worked across Canada.

6.5. Discussion

The results we have uncovered on the labour force participation of mothers (along with previous researchers on the topic) are large, significant and persistent. This is unique in the literature. The labour force participation decision of mothers is particularly difficult to analyse because the decision is sensitive not only to the cost of childcare and the age of the children covered but also to the quality, availability, convenience, reliability and security of childcare. We have already established that the cost of childcare was extremely low in Québec, which certainly helped trigger the labour supply effect. But most likely it's the combination of low cost and all the other characteristics that led to this dramatic change. With a growing number of spaces over time covering a large percentage of children all across Québec, and with each space providing parents with a guaranteed service for a full year (261 days) and a full day of 11–12 hours, the service provided by the program was, and is to this day, considered highly available, convenient and reliable.[23] Regarding quality, we have to make a distinction between measured quality and perceived quality. With a common government-regulated educational program and no explicit measures of quality at the time of the implementation, parents were often not able to form a qualified judgment about quality. As a result, perceived quality was generally high in the early years of the program such that low quality was certainly not a deterrent for choosing a 'low-fee' space.

Finally, in the 2000s the Canadian economy outperformed the American economy (Macdonald, 2007). The unemployment rate fell over the decade with substantial job growth. In Québec, the economy followed similar trends, with a gap in annual growth of 0.5% entirely explained by a lower growth rate in the population aged 15–64 years old. Some recent estimates of childcare subsidies on labour supply in other countries show smaller effects than those found in research studies on the Québec reform. We believe that the main explanations are the differences in service offering and in labour market conditions. During the time span we observe, from 1994–2008, the North American labour market was generally booming, with jobs being created at an extraordinary rate. The Québec policy was implemented as labour demand was rapidly increasing. The newly available labour supply created by the policy was easily absorbed by the labour market. Also, a large majority of young mothers in Canada were rather skilled. Québec was no exception, with a high proportion of young mothers

[21] See Fig. A.2 in the Appendix A for more details.

[22] These estimates can be obtained from the authors on request.

[23] Newspapers often report a lack of low-fee spaces, but it is mainly a shortage of low-fee spaces in not-for-profit centres that dominates the discussion.

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Table 5
Estimated effects of the policy on mothers' labour force participation and weeks worked by the mother's level of education.

Age of children	1	2	3	4	5 not eligible for K-5	5 eligible for K-5	1–4
Labour force participation							
Some postsecondary or less							
β_3 (1998–99)	0.04	0.07	0.01	−0.06	−0.19**	−0.09	0.02
	(0.04)	(0.08)	(0.06)	(0.07)	(0.09)	(0.05)	(0.02)
β_4 (2000–01)	0.05	0.06	−0.01	0.06	−0.14	−0.05	0.04
	(0.06)	(0.06)	(0.05)	(0.07)	(0.10)	(0.07)	(0.03)
β_5 (2002–03)	0.12	0.06	0.04	0.00	−0.04	−0.01	0.05
	(0.07)	(0.07)	(0.06)	(0.06)	(0.10)	(0.06)	(0.03)
β_6 (2004–05)	0.13*	0.22***	0.13*	0.08	−0.10	0.10	0.14***
	(0.07)	(0.07)	(0.08)	(0.08)	(0.14)	(0.08)	(0.04)
β_7 (2006–07)	0.14*	0.15*	0.05	0.04	0.18	0.13	0.09**
	(0.07)	(0.08)	(0.07)	(0.07)	(0.18)	(0.08)	(0.04)
β_8 (2008–09)	0.08	0.20**	0.11	0.00	−0.09	0.00	0.10***
	(0.07)	(0.09)	(0.07)	(0.09)	(0.15)	(0.09)	(0.04)
N	10,194	7215	9260	7080	3556	8592	33,749
Postsecondary diploma or more							
β_3 (1998–99)	0.04	−0.05	−0.03	−0.04	−0.05	0.04	−0.02
	(0.04)	(0.07)	(0.06)	(0.06)	(0.09)	(0.06)	(0.03)
β_4 (2000–01)	0.11**	0.10**	0.04	−0.03	−0.10	0.03	0.06*
	(0.05)	(0.05)	(0.05)	(0.08)	(0.10)	(0.07)	(0.03)
β_5 (2002–03)	0.15***	0.07	0.07	0.06	0.06	0.02	0.10***
	(0.05)	(0.06)	(0.07)	(0.06)	(0.09)	(0.07)	(0.03)
β_6 (2004–05)	0.14***	0.04	0.16***	0.07	0.04	0.13*	0.10***
	(0.05)	(0.06)	(0.06)	(0.06)	(0.12)	(0.07)	(0.03)
β_7 (2006–07)	0.11**	0.05	0.20***	0.00	0.11	0.13*	0.09***
	(0.05)	(0.07)	(0.05)	(0.07)	(0.10)	(0.08)	(0.03)
β_8 (2008–09)	0.04	0.14**	0.10*	0.13**	0.08	0.16**	0.11***
	(0.05)	(0.06)	(0.05)	(0.06)	(0.1)	(0.07)	(0.03)
N	9456	6899	8950	6826	3112	8016	32,131
Weeks worked							
Some postsecondary or less							
β_3 (1998–99)	1.17	3.19	1.28	−3.88	−7.57	−3.74	0.36
	(2.06)	(3.45)	(2.81)	(2.85)	(4.63)	(2.67)	(1.46)
β_4 (2000–01)	0.41	2.06	−0.50	2.44	−5.11	−3.85	1.02
	(2.74)	(2.59)	(2.42)	(3.55)	(5.00)	(3.24)	(1.56)
β_5 (2002–03)	4.27	2.47	0.64	−0.05	0.73	−2.48	1.68
	(3.31)	(3.30)	(3.09)	(2.95)	(4.76)	(3.13)	(1.57)
β_6 (2004–05)	4.12	9.28***	4.21	5.63	1.00	3.86	5.71***
	(3.36)	(3.24)	(3.52)	(3.87)	(6.68)	(3.85)	(1.83)
β_7 (2006–07)	4.38	8.64**	3.36	5.69	8.30	2.85	5.49***
	(3.23)	(4.22)	(3.50)	(3.70)	(8.42)	(4.07)	(1.83)
β_8 (2008–09)	3.95	5.95	2.52	3.13	−4.80	−0.03	3.63*
	(3.21)	(5.03)	(3.23)	(4.74)	(7.35)	(4.49)	(2.05)
N	10,150	7208	9253	7091	3555	8579	33,702
Postsecondary diploma or more							
β_3 (1998–99)	2.81	−2.82	−2.29	−1.67	1.27	−0.34	−1.38
	(2.22)	(3.35)	(3.07)	(3.14)	(4.76)	(3.42)	(1.58)
β_4 (2000–01)	5.67**	1.86	0.87	−0.18	−4.77	0.52	2.03
	(2.84)	(2.70)	(2.55)	(3.80)	(5.55)	(3.79)	(1.60)
β_5 (2002–03)	10.94***	1.86	3.21	6.95**	4.34	0.51	5.99***
	(2.95)	(3.25)	(3.61)	(3.08)	(5.19)	(3.64)	(1.71)
β_6 (2004–05)	6.72**	1.20	10.13***	6.50**	5.24	3.81	5.99***
	(2.79)	(3.26)	(3.09)	(3.3)	(6.8)	(4.02)	(1.65)
β_7 (2006–07)	3.57	2.03	9.70***	0.98	8.15	6.62	4.13**
	(2.82)	(3.52)	(2.90)	(3.63)	(6.46)	(4.11)	(1.75)
β_8 (2008–09)	5.12*	8.05**	5.80**	8.08**	10.07*	5.91	6.91***
	(2.78)	(3.31)	(2.84)	(3.19)	(5.41)	(3.96)	(1.66)
N	9435	6893	8949	6827	3118	8010	32,104

Note: The dependent variables are a dummy for labour force participation of mothers with some postsecondary education or less (panel 1) and mothers with at least a postsecondary diploma (panel 2), and the number of weeks worked for mothers with some postsecondary education or less (panel 3) and mothers with at least a postsecondary diploma (panel 4). Children from all family types are included. We include the same controls as in Table 2. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is $p < 0.01$, ** is $p < 0.05$, and * is $p < 0.1$.

with a postsecondary diploma. Therefore, not only did mothers receive the support of the daycare network to allow them to work, but also the economy was able to absorb rather rapidly this large shift in labour supply. During our sample period, Continental European economies had less dynamic labour markets due to institutional constraints such as high costs of dismissal and high payroll taxes, in particular for small businesses (Garibaldi and Mauro, 2000). This may explain lower effects of certain childcare policies in Europe. Also, in Québec, participation rates for mothers with young children were relatively low compared to those in the RofC prior to the reform. Therefore, there was considerable room to move up. The flattening out of effects in the later years could be due to the fact that mothers who are now not in the labour market are mothers with very low skills or very strong preferences for staying home with young children.

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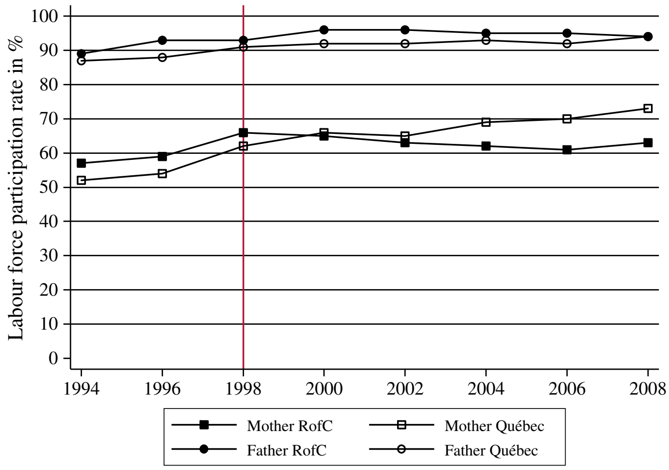

Fig. 5. Labour force participation of mothers (two-parent and single-parent families) and fathers by region. Note: Shows the evolution of paternal labour force participation for the NLSCY cross-sectional children aged 1–4.

As discussed above, specific Québec policies may interact with the daycare reform. One policy in particular caught our attention: the new child benefit program introduced in July 1997, which replaced a benefit based on the age of the child with a benefit contingent on income. Practically, the policy increased child benefits for families with less than 26,000 dollars while reducing those for families with more than 26,000 dollars in yearly income. For low-income families (less than 26,000 dollars), the policy increased financial assistance by 726 dollars annually on average, while for higher income families, there was a decrease in benefits of 600 dollars or less. For low-income working single mothers, financial assistance increased by about 1000 dollars. For single mothers on welfare, the incentive to work may therefore have seemed large at first glance. However, a detailed study by Rose (1997), taking into account all transfers and work-incentive programs for welfare mothers at the time, showed that the new net incentives were small at less than \$500. Our estimated impacts on labour force participation for less-educated mothers in the early years of the daycare reform (when the number of spaces remained low and only 4-year-olds were eligible) are not statistically different from zero. This suggests that not only did the daycare reform have no impact on labour force participation of less-educated mothers early on, neither did the new child benefit program. We are therefore confident that the effects we uncovered later were due exclusively to the daycare reform and not to the Québec child benefit reform, which changed in July 1997 and remained constant thereafter.

Finally, in 2005, an earned income tax credit was added for low earners and all types of family aid were regrouped into one family benefit. We do not observe any major changes in our estimated effects of the daycare reform after 2005 for less-educated mothers and their children: the estimated effects prior to 2006 were already large and remained equally large afterwards. The positive effects we uncover beyond 2005 are most likely due to the Québec childcare policy.

Table 6
Estimated effects of the policy on fathers' labour force participation and weeks worked by age of the children.

Age of children	1	2	3	4	5 not eligible for K-5	5 eligible for K-5
Fathers' current labour force participation						
β_3 (1998–99)	0.01	0.02	−0.04	0.00	−0.07*	−0.05*
	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)	(0.03)
β_4 (2000–01)	0.03	0.01	0.02	0.01	−0.01	−0.01
	(0.03)	(0.03)	(0.02)	(0.04)	(0.05)	(0.03)
β_5 (2002–03)	0.04	−0.01	0.02	−0.01	−0.02	−0.02
	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	(0.03)
β_6 (2004–05)	0.06**	0.03	0.05**	−0.04	−0.06	−0.01
	(0.03)	(0.03)	(0.02)	(0.04)	(0.07)	(0.04)
β_7 (2006–07)	0.02	−0.03	0.03	0.02	−0.11	−0.01
	(0.03)	(0.04)	(0.02)	(0.03)	(0.10)	(0.03)
β_8 (2008–09)	0.01	0.01	0.02	0.03	−0.08	0.02
	(0.03)	(0.03)	(0.03)	(0.03)	(0.06)	(0.03)
N	16,984	12,108	15,518	11,760	5623	13,855
Fathers' weeks worked in reference year						
β_3 (1998–99)	−0.00	−1.04	−1.89	0.42	−1.99	−2.87**
	(1.24)	(1.50)	(1.31)	(1.25)	(1.73)	(1.22)
β_4 (2000–01)	2.20*	−0.44	0.54	0.49	−1.58	−0.29
	(1.23)	(1.32)	(1.25)	(1.45)	(2.11)	(1.38)
β_5 (2002–03)	1.30	−0.47	0.48	0.51	−0.60	−1.39
	(1.46)	(1.46)	(1.28)	(1.31)	(2.05)	(1.58)
β_6 (2004–05)	1.32	−1.35	2.48**	−3.38*	−2.77	−0.33
	(1.21)	(1.45)	(1.10)	(1.88)	(3.09)	(1.70)
β_7 (2006–07)	−0.33	−3.40*	0.66	−0.01	−4.76	−1.59
	(1.35)	(2.01)	(1.22)	(1.51)	(5.01)	(1.91)
β_8 (2008–09)	0.13	0.64	−0.44	0.43	−4.10	−1.08
	(1.32)	(1.45)	(1.34)	(1.40)	(2.88)	(1.58)
N	16,916	12,084	15,495	11,759	5618	13,889

Note: The dependent variables are a dummy for paternal labour force participation (top panel) and the number of weeks worked by the father in the reference year (bottom panel). Children from all family types are included. We include the same controls as in Table 2. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is $p < 0.01$, ** is $p < 0.05$, and * is $p < 0.1$.

7. Cognitive development

Using the NLSCY data on 4- and 5-year-olds not yet eligible for kindergarten, we estimate the impact of the reform on the Peabody Picture Vocabulary Test (PPVT). We first provide a brief overview of the test and then discuss the results.

7.1. Peabody Picture Vocabulary Test (PPVT)

The PPVT is a widely used measure in the literature on early childhood development (Mayer, 1997). It measures the listening comprehension for spoken words in standard English (French for children with French as their mother tongue) and may be used to measure school readiness for children in these age groups. More specifically, for the PPVT, children are presented with a number of pictures arranged in a multiple-choice format and asked to select the one that best illustrates the meaning of the word orally presented by the examiner. Vocabulary is highly correlated with general intelligence and represents knowledge closely aligned with the 'cultural capital' of the child's environment. It is not easy to increase vocabulary knowledge as it depends on years of conversations between adults and children; and it also depends on adults reading to children.

The NLSCY master files provide both the PPVT raw (PPVT-Raw) and the PPVT standardised (PPVR-SD) scores. Statistics Canada has used different methodologies to standardise the scores over time. The released PPVT-SD scores for waves 1–3 are standardised within wave (with slight variations in the methodology), while for waves 4–8 scores are standardised over the grand population of all tests over the first 5 waves of data. In waves 1–3, scores within wave are standardised by age group to have the same mean of 100 and standard deviation of 15. This type of standardization is common for analysis of domains within a wave; however it provides limited insight for in-between wave analysis. For robustness and to allow analysis of changes through time (to capture true population differences over time and not simply differences resulting from sampling error), we have restandardised (using Statistics Canada smoothing routine) the PPVT-Raw scores using all 8 waves of the NLSCY (as of now PPVT will refer to this standardised score). [24]

[24] Following the approach of Statistics Canada, the standardization was done separately for the PPVT and the EVIP (the acronym for the French adaptation of the test). This should be of no concern since our estimates are based on differences in scores over time between Québec children (80% of whom speak French) and those from the RofC or Ontario.

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We selected all children aged 4–5 years old from the data sets of all 8 waves.

Note that this section is not about the evaluation of the effects of childcare on the development of children, but an evaluation of the effects of a low-fee for long-hours daycare policy on child development. Because most children who are in daycare have working mothers, regression methods have difficulty disentangling the effects of nonparental daycare relative to parental daycare from the effects of having a working mother relative to a mother staying at home. Second, because using childcare is a choice, it is a function of unobservable preferences that can also determine the cognitive performance of children. For example, mothers who stay home with their child may, in general, prefer spending long hours with their child, which could be beneficial to the child. On the other hand, some mothers with very low levels of human capital do not work and do not use childcare, and their children score poorly on cognitive tests on average. These types of biases are not present in our analysis given that we are estimating the impact of the policy and not daycare attendance. Again, we estimate the model presented in Eq. (1). Note that consistent estimates of the effects of daycare and hours of work on child development were obtained by Bernal (2008) with a structural model and instrumental variables as well as credible assumptions. Bernal's paper showed the complexities of studies that seek to obtain structural estimates in this area of research.

The estimated effects on the PPVT score are presented in Table 7.[25] For the full sample of children aged 4–5 years old and not yet eligible for school,[26] we find that the effects measured by the PPVT (column 1) are negative, but generally not significant. When we separate the sample by the education level of the mothers, the negative effects become much larger for children of low-educated mothers (column 2), reaching a significant level of −6.02 in 2004 and −4.89 in 2008 (a third of a standard deviation). Strangely, the 2006 estimate for this group is negative but not significant, however there are only 103 observations in Québec in 2006. The coefficients for children with a postsecondary-educated mother (column 3) is usually small, negative and never significant. Children in Ontario provide an interesting comparison group as the province offers half-day kindergarten to 4-year-olds. When they are used as a control group, we find again nonsignificant effects that are somewhat smaller in magnitude (column 4). The signs generally suggest negative effects, stronger for children of low-educated mothers. Clearly, the policy has no positive effects on the cognitive development of children aged 4–5 years old measured using the PPVT. One explanation might lie in the increasing number of hours children spend in daycare. When looking at the cumulative number of hours spent in care over time, we find that it, indeed, increases over time (Table A.3).

In sum, the results are not significant for children aged 4–5 years old, except for children with the less-educated mothers of our sample. Since one of the objectives of the policy was to increase early literacy skills and better prepare children for school, our results suggest that on this front the policy may not have been successful.

Table 7
Estimated effects of the policy on PPVT scores.

4–5 years old not eligible for K-5				
All Children		Some Postsec or Less	Postsec Diploma or More	Ontario
PPVT				
β_3 (1998–99)	−0.63	0.28	−1.58	−1.62
	(1.16)	(1.51)	(1.74)	(1.36)
β_4 (2000–01)	−1.99	−2.39	−1.28	−0.37
	(1.41)	(1.73)	(2.22)	(1.68)
β_5 (2002–03)	−1.54	−2.35	−0.28	−1.31
	(1.22)	(1.49)	(1.93)	(1.39)
β_6 (2004–05)	−3.86***	−6.02***	−1.86	−3.02*
	(1.41)	(1.96)	(2.02)	(1.61)
β_7 (2006–07)	−1.41	−1.89	−1.10	−1.64
	(1.52)	(2.05)	(2.11)	(1.77)
β_8 (2008–09)	−1.39	−4.89*	−0.20	−0.78
	(1.68)	(2.70)	(2.19)	(1.87)
N	18 026	9 135	8 891	7 739

Note: The dependent variable is the PPVT standardized score. The subsamples are identified at the top of each columns: all NLSCY cross-sectional children (All), children of mothers holding no more than some postsecondary education (Some Postsec or Less), children of mothers holding at least a postsecondary diploma (Postsec Diploma or More) and all children of Québec versus all of Ontario (Ontario). We include the same controls as in Table 2. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is $p < 0.01$, ** is $p < 0.05$, and * is $p < 0.1$.

7.2. Discussion

In summary, the results by mothers' education level suggest that the policy did not reduce 'social' gaps in school readiness. In fact, the negative estimated impacts of the policy are generally larger on average for a child whose mother is less educated. This is consistent with the previous work of Baker et al. (2008), which found a negative impact of the policy on several child behavioural indicators such as hyperactivity. They did not find a negative impact on child development at age 4, but their data set spanned only the first four years of the policy and did not provide any estimates by the mother's education.

We provide two major explanations to help reconcile our results with former studies on the impact of ECEC on preschool cognitive development. First, rarely can we observe variations in hours of non-parental care for young children of the magnitude observed after the implementation of the program. Services are available 10–12 hours (depending on the setting) per day, 261 days a year and at the same low-fee per day for all children. For example, using the NLSCY, we observe that in 1994, 45% of all children aged 1–4 years old were in childcare and 68% were in childcare for more than 21 hours per week (excluding children with 0 hours); in 2002, 70% of children aged 1–4 years old were in childcare and 78% were in childcare for more than 21 hours; in 2008, 78% were in childcare and 83% were in childcare for more than 21 hours per week. Not only were more children in daycare, but they were there for much longer hours. The increase in the number of hours certainly reflects the increased labour force participation of mothers, but it may also be attributed to the structure of the program and its financial incentives.[27] The government asks daycare providers to ensure that parents use daycare services every day of the week (unless the child is sick or on vacation with the family): if a space is not occupied full-time the subsidy may be reduced. In theory, a provider may offer part-time spaces, but in practice they almost exclusively offer full-time full-week spaces because it is easier to manage. The last table in the Appendix A shows the policy impacts on the sum of hours in childcare in the surveys for longitudinal children (who are surveyed for 3 cycles: for one group at ages 0, 2, and 4 years old; for the other group at ages 1, 3, and 5 years old. For children aged 1–5 years old who were not in school, the effect is 16 hours in cycle 8. This shows that Québec children in 2008 spent much more time in childcare since birth than did children in the RofC. As referenced in the Appendix A, Belsky et al. (2007)

[25] Estimates using the PPVT raw score while controlling for age at the time of test lead to comparable conclusions and can be obtained from the authors on request.

[26] This is determined using the birth date of the child and provincial regulations regarding school entry age.

[27] In fact, the $7 day fee translates to $3.5 (approximately 3 euros at the actual 2013 exchange rate) after taking into account the federal income taxes (childcare fees are deductible expenses), family transfers based on net taxable income, and a universal federal subsidy of $100 per month for each child aged 0–5 years old to help families pay for childcare.

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Table 8
Parental work-childcare profile and mode of care used by quartile of gross family total income.

	Quartile 1	Quartile 2	Quartile 3	Quartile 4	All
Family total income quartile cut-off point	$47,400	$71,200	$102,220	-	na
Mean income of quartile	$29,744	$59,101	$84,943	$140,208	$78,447
Parental work-childcare usage profile					
Parent(s) work/study and use childcare	51	67	73	84	69
Parent(s) work/study and no childcare	14	12	16	8	12
Parent(s) do not work/study	36	21	11	8	19
Total	100	100	100	100	100
Mode of care used					
Family-based (other home)	8	11	13	8	10
Family-based (other, relative)	5	3	5	2	4
Child's home (by a non relative/relative)	7	6	3	8	6
Daycare centre	31	45	52	65	47
Does not use childcare (not working/studying)	45	31	25	14	30
Not asked	4	4	2	3	3
Total	100	100	100	100	100
Number of children un-weighted	372	400	401	315	1482
Weighted observations	101,763	101,312	101,466	101,299	405,840

Note: The statistics presented in this table were computed using the 2008–2009 NLSCY data on cross-sectional children aged 0–59 months old in Québec. The mode of care used is only asked to parents using childcare for work or study. As such, we pooled children of parents who do not work or study under the label "Not asked".

presented convincing evidence that long hours in daycare can be detrimental to child development.

Second, although more children are now in regulated types of daycare – which is supposed to be helpful or at a minimum not harmful, in particular for children from disadvantaged households – two major studies on the first years of the program (ISQ, 2004; Japel et al., 2005) showed that the average quality in Québec's subsidised daycare network was at best satisfactory and in many cases low or not acceptable, particularly for children in lower-income families. According to Japel et al. (2005), more than 20% of children from the lowest income quartile in daycare were in daycare of inadequate quality compared to 10% for the two upper quartiles. Approximately 70% of lower income children in daycare were in daycare of average quality or lower. Part of this is explained by the rush to implement the program, build up new settings and create new spaces to respond to the excess demand for spaces, which forced the government to recruit daycare workers with no specific training in ECEC. The Ministry of the Family imposes an 'official' ECEC program (embedded in the childcare law) that childcare providers must follow. However, according to a recent audit on educational childcare services conducted by the Office of the Auditor General of Québec (VGQ, 2011), it is difficult to determine if the program is implemented in specific daycare settings, and no action to enforce the program or measures to train the educators have been observed across the province. The audit also finds that the percentage of subsidised centres not respecting the maximum ratio of number of children per qualified educator was 42% during 2008–2009 and 54% during 2009–2010. Finally, as discussed above, Table 8 shows that quartiles of income are highly correlated with the more formal mode of care (i.e., daycare centre). As these centres generally offer a higher quality of service, this evidence further supports the idea that low quality likely explains the lack of impact of the policy on cognitive development and could lead to poorer results for children from low-income families.

In light of this evidence, it is important to stress that the negative effects we uncovered may not relate to childcare per se, but to the structure of the program that created strong incentives for families to use long hours of care starting at a very young age and offered at the best average quality. A number of policies possibly influencing child development were mentioned in Section 4. Reforms common to both Québec and the RofC are accounted for through our control group. However, the impact of the Québec 2005 earned income supplement remains unaccounted for. It benefited families by raising their disposable income. This likely has positive impacts on child development, or at least no impact. This implies that the negative effects we have uncovered with respect to the daycare reform are most likely underestimated as of 2006.

The results obtained are consistent with those of Bernal (2008), who found negative effects of increased labour force participation and childcare use on child development in a structural model carefully specified to avoid the usual biases in this field of research. For children whose mother's behaviour changes from no work and no childcare use to full time work and childcare use for a whole year, the estimates are that average ability will decrease by 0.13 of a standard deviation. The effect of the policy on the PPVT scores for children aged 4–5 years old in cycle 8 in our study is of the order of −1.3, which is about 0.09 standard deviation; however, for children of low-educated mothers the effect is larger at −4.8 (about one-third of a standard deviation). This corresponds to children in families having benefited from the policy since birth (aged 1– 4 years old). However, we are not comparing children of full-time working mothers who use childcare with children with mothers who do not work and do not use childcare, but 4- and 5-year-old children before the policy and those eligible for low-fee daycare since birth. Therefore, even though children now have mothers with much higher participation rates who almost all use childcare, the effect on children of low-educated mothers is relatively large compared to the effect estimated by Bernal and Keane (2011) on single mothers (a decrease of 0.44 standard deviation over 4 years). If we add negative effects that could be due to a high intensity of care as well as quality of care problems, the size of these effects become more realistic. Japel et al. (2005) showed that almost 20% of children in the first two income quartiles were in low-quality childcare settings. This is found in a study where the response rate of providers was 50%. If we assume that childcare facilities of poor quality are those that refused evaluation and that a large proportion of poor children are in these facilities, it is possible that the percentage of low-income children in childcare of inadequate quality is even higher.

7.3. Lessons learned

It is useful to point out the lessons that can be learned for policy-makers who seek to increase maternal labour supply and improve school readiness by increasing childcare subsidies. First, the approach that is based on a very low fixed price and direct subsidies to childcare providers combined with a large expansion in the number of available spaces for

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small children likely explains part of the effects on labour supply. Second, families with liquidity constraints, in addition to receiving a low price for care, are better off in a setting where the upfront cost is low compared with a high upfront tax-deductible cost. Third, in general, suppliers provide long hours and are always open, which allows mothers who need more flexibility and reliability for childcare to accept jobs with longer hours.

However, there are several costs to this policy, over and above subsidies to providers. First and foremost is the unionisation of childcare workers that greatly increased the hourly wage and pensions, which ultimately strained the ability of the government to increase resources and offer better quality, including the recruitment of highly qualified workers for whom the unionised wage remains too low. Additional costs are linked to the creation of low-fee spaces as the government participated in the financing of new infrastructure. The most disturbing aspect of this policy is that by creating new spaces progressively at the beginning, a large number of parents were left out during the first few years. The lack of supply frequently made headlines in the news and eventually forced the government to create spaces at a very rapid rate, with an evident lack of trained personnel. The lack of training may in part explain the negative effects we find on school readiness.

Finally, different modes of care are available to families as a result of the policy, and important variations in care arrangements are observed by income group. Using the NLSCY wave 8 (2008–2009) data, we document the working status of parents and childcare use by family total current-income quartile, as well as the percentage of children aged 0–59 months old in different modes of care by family income quartile. Table 8 shows that quartiles of income are highly correlated with the more formal mode of care (i.e., daycare centre). Approximately 65% of children with parents in the highest income quartile attend daycare centres, compared to 31% in the lowest income quartile. A high proportion of low-income families (quartile 1) do not use childcare at all (45%). Of those who work or study, only 8% do not use childcare in the highest income quartile, compared to 14% in the lowest income quartile. These figures show that there are important differences in care arrangements across the income distribution. For fiscal year 2011–2012, the total yearly subsidy for a child aged 0–17 months old in a not-for-profit centre is $17,913, and for a child aged 18–59 months old is $12,406. The average subsidies in a for-profit centre for children of the same age (as above) are, respectively, $16,037 and $11,456. The subsidy for family-based childcare is on average $9000. As a result, high-income families receive (indirectly, not taking into account that they pay higher income taxes) a disproportionate share of the total subsidy to the program not only because they use childcare services more, but also because they use the more expensive type of childcare services. As not-for-profit centres generally offer a higher quality[28] of service, these findings raise the question of vertical equity, both in terms of quality and quantity of services received.

A more targeted policy of this type in low-income neighbourhoods would have probably had a better chance of helping children in need of educational services. First, the costs would have been lower. Second, the political pressure to create spaces would have been much lower, making the implementation smoother. Third, more money would have been available for quality purposes. Some type of incentive mechanism to reduce the number of hours spent in daycare should have been implemented as this may also have affected child development.

8. Conclusion

In this paper we analyse the long-term effects of the Québec daycare reform. We showed that the reform had important and lasting effects on the number of hours children spent in daycare from age 1–4 years old. Not only did the number of children attending daycare increase, but also the number of hours they spent in daycare conditional on attending daycare.

The NLSCY compared to the SLID provided sufficient data to estimate the effects of the policy by subgroup. We found that the increased labour force participation and weeks worked of mothers was also long lasting and mainly driven by highly educated mothers. Interestingly, for children aged 5 years old, we uncovered strong evidence that implementing full-day kindergarten alone was not enough to increase maternal labour force participation and weeks worked, but when combined with the low-fee daycare program it was. Indeed, mothers whose child was eligible for low-fee daycare early on were more likely to participate in the labour market once their child reached age 5 years old, compared to mothers whose child was not. The policy did not have an impact on the labour force participation of fathers, and had minor impacts on their number of weeks worked.

Finally, but certainly not least importantly, we found negative impacts on the cognitive development of the 4–5-year-old children of low-educated mothers. Clearly, the daycare reform did not raise the average cognitive ability of these children. Again, we emphasise that we are estimating the effects of a complex daycare policy, which increased the number of hours of care and offered at best average quality care, as opposed to the effect of childcare per se. Raising the qualification of childcare educators or a junior kindergarten program for the 4-year-olds, in particular those of disadvantaged households, could be helpful for the cognitive development of such children.

Appendix A

In this Appendix we present additional information on the evolution of the childcare network and the net cost to families, and the literature on ECEC and child development. Additional figures and tables are also presented to further support our paper's discussion.

A.1. Evolution of the childcare network and net costs to families

Fig. A.1. shows the evolution in the total number of spaces. This figure also shows that as of 2000, not-for-profit centres and family daycare services offered exclusively low-fee spaces and continued to do so over the entire observation period. For-profit centres provide a fairly stable number of spaces over the period, except in recent years. These centres regroup those that are subsidised by the government (and offer low-fee spaces) and those that are fully private but regulated. Prior to 2001, these two categories cannot be distinguished with publicly available data. As of 2001, most of these centres were subsidised by the government and offered low-fee spaces (97%). While the number of low-fee spaces in for-profit centres remained stable from 2001 onwards, the number of fully private regulated spaces increased.

Even if the number of spaces has been steadily increasing, the perception has always been that the demand for low-fee spaces was above the number of available spaces because of long waiting lists. Unregulated spaces eligible for a fiscal treatment (a deduction at the federal level and a refundable tax credit at the provincial level) offer an alternative solution to regulated spaces. To ease off the pressure on the demand for low-fee spaces and to reduce the net-cost difference between private childcare and low-fee childcare, the scale of the refundable tax credit for those not using low-fee daycare was increased twice (2008 and 2009). Since 2009, the maximum annual expense eligible for a tax credit is $9000 per child less than 7-year-old. The minimum credit rate is 44% (compared to 26% earlier) and the maximum remained unchanged at 75%. The minimum rate is applied to a family whose household income is $125,000 or over (compared to $84,040 earlier). In 2004, the daily price of $5 was increased to $7.

[28] The ISQ report showed that a higher proportion of not-for-profit centres offered good or very good quality, more than 40%, compared to only 19% in family-based care and 11% in for-profit centres.

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Accounting for federal tax credits and family transfers, the net cost for a subsidised space for a child under age 5 years old is in fact $2.87 per day for a family income up to $150,000. Assuming a chilcare cost of $25 per day per child for 260 days ($6500), the enhanced refundable tax credit reduces the net cost to $3.31 per day for families with less than $125,000 of income. This partially explains the recent increase in the number of spaces in for-profit centres (see Fig. 1).

A.2. Complement to ECEC and child development literature

In this section we expand some of the content presented in the paper while keeping the same structure (and in some cases we repeat some of the content of the paper). Again, for a more detailed overview of the literature we refer the reader to Almond and Currie (2011). We first briefly review research on children aged 0–2 years old, then on children aged 3–5 years old. Then, we discuss evidence on the impact of the quality and intensity of care, and the optimal entry age in daycare. We provide additional details on research on part-time versus full-time kindergarten. Finally, we provide further details on the measures used by Baker et al. (2008).

First, on children aged 0–2 years old, there is a growing body of empirical results indicating that maternal employment and time spent in childcare during the first year of life can have adverse effects on a child's developmental outcomes (such as verbal, reading and math scores, and indices of behavioural problems) observed at later ages (Rhum, 2004; Waldfogel et al., 2002; Hill et al., 2005), even after controlling for childcare quality, the quality of the home environment, and maternal sensitivity (Brooks-Gunn et al., 2002; Hill et al., 2002; and for United Kingdom, Gregg et al., 2005).

Second, for children aged 3–5 years old, several studies on the effect of preschool programs find significant positive effects on cognitive outcomes (letter-word identification, spelling and applied problems) and measures of school readiness (Gormley and Gayer, 2005; Gormley et al., 2005; Magnuson et al., 2004, 2007). For disadvantaged children (whether defined by poverty status, low maternal education, single-parent headship, or mothers who do not speak English) the cognitive gains are larger and longer lasting (e.g. Burger, 2010) A large-scale UK study following children aged 2 years old or more attending centre-based preschool finds similar results (Sammons et al., 2002, 2003).

Third, while the quality of care seems to have positive, but small, effects on cognitive outcomes (Blau, 1999; Duncan and NICHD Early Child Care Research Network, 2003; Belsky et al., 2007), longer hours in all types of preschool programs seem to be associated with increased behavioural problems that persist over time (Magnuson et al., 2004, 2007). On the intensity of care, Loeb et al. (2007) find that for disadvantaged children (aged 2–3 years old), 30 hours per week of care for at least 9 months per year has little detrimental effect on their behaviour, while producing positive effects on their cognitive outcomes. In contrast, the cognitive development of children from wealthier households appears to benefit from daycare outside the home only if it is part-time (between 15–30 hours per week for at least 9 months per year).

Fourth, the existing American studies on full-day versus half-day kindergarten suggest that the impact of full-day kindergarten on academic and social outcomes is somewhat mixed. Most studies fail to control for student assignment to kindergarten programs and have fairly short observation periods. Using American longitudinal data providing extensive information on the child's socioeconomic context and school[29] Lee et al. (2006) find that full-day kindergarten children learn more in literacy and mathematics over the kindergarten year than those in half-day programs. In contrast to previous studies, they do not find that full-day kindergarten is more effective for low-income/at-risk children. Using data from the same survey, DeCicca (2007) finds that full-day kindergarten has sizeable impacts on academic achievement during the first year of program implementation, but the estimated gains are short-lived, particularly for minority children. However, the efficiency of the full-day kindergarten intervention may be contingent upon class size as students in smaller full-day classes benefit more (Zvoch et al, 2008).

Finally, in Canada, the Baker et al. (2008) study highlights some negative effects of the Québec daycare policy on both behavioural scores and health indicators of Québec's children and parents. For the 0–2 years old, parent-reported measures such as "never had a nose/throat/ear infection" are used. These are benign health outcomes and this life experience may have reinforce their immunitary system and their long term health (Ball et al. 2002; Lu et al., 2004; Côté et al., 2010). For children aged 2–3 years old the behaviour of the child is measured using an emotional anxiety score, a physical aggression score, and a separation anxiety score, each based on a serie of parent-reported measures. Finally, for the 0–4-year-olds, the authors use the answer excellent to the question: "In general, would you say this child's health is … at time of survey". For parents, the mother's depression score and the family dysfunction index are used.

A.3. Cost-benefits/Fiscal benefits of the policy

We now turn to the fiscal benefits of the childcare policy. By increasing the labour supply of mothers with young children the policy has increased the tax base for both the provincial and federal governments. Also, given that tax credits decrease with 'net' family (taxable) income, both federal and provincial transfers should be lower as a result of the policy. We simulated the fiscal benefits of the policy for the year 2004. This year is chosen because the program was fully implemented and the year after (in 2005) an earned income tax credit policy for low-income families participating the labour market was introduced in Québec.[30] We use Statistics Canada's SLID data set which provides information on income sources, taxes, credits, and transfers for an individual and for the family. In line with the results of Lefebvre et al. (2009) based on the SLID, we suppose that the policy increased the participation rate of mothers who have at least one child aged 1–11 by 10 percentage points.

To compute the fiscal impacts we use the software written by Milligan (2008). We first simulate the income taxes and transfers for all working mothers and then subtract the simulated income taxes

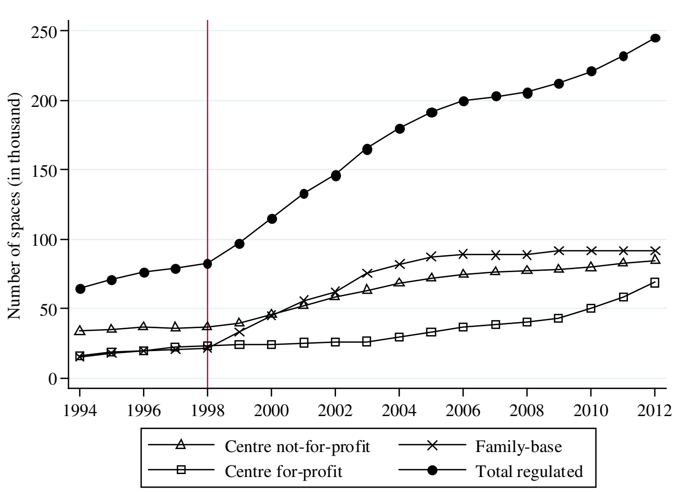

Fig. A.1. Number of regulated spaces.

[29] The Early Childhood Longitudinal Study – Kindergarten cohort, a nationally representative sample of over 8000 kindergarteners.

[30] More specifically, as of January 2005, the wage subsidy replaced an extremely targeted wage assistance program. Also at the same time, the non-refundable tax credit for dependent children, the family allowances and the tax reduction for families were replaced by a single enhanced child tax benefit.

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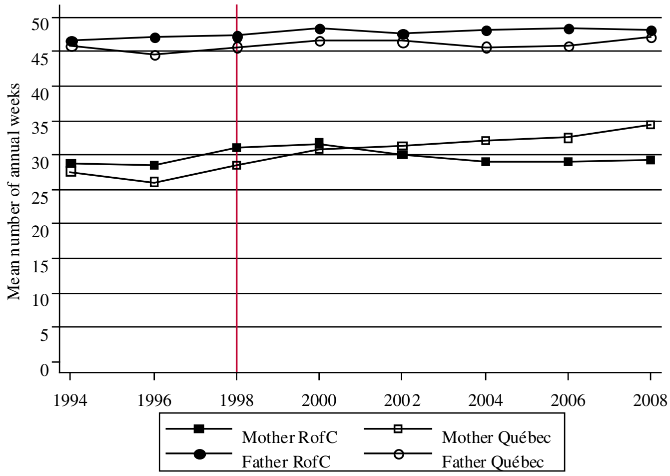

Fig. A.2. Annual weeks worked by mothers (in two-parent or single-parent households), fathers and families by region. Note: Shows the evolution of parental weeks worked for NLSCY cross-sectional children aged 1–4.

and transfers assuming that 10% of mothers leave the labour market (corresponding to the counterfactual no daycare policy). The question then becomes which mothers' earnings are set to zero? The benefits to the government of increased labour force participation are most likely a positive function of the earnings of the mothers reacting to the policy change. To estimate the benefits of the daycare program to the government, we perform a number of simulations by changing the labour force participation of different subsets of mothers. This allows us to bound the benefits to the government.

Effectively, we first order mothers based on their earnings, while keeping only mothers with positive earnings in our data set. Taking into account the population weight of each mother, we set the earnings to 0 for a number of earners that correspond to 10 percent of our sample (reducing the participation rate by 10 percentage points, this corresponds of course to more than 10% of earners), and we run our first simulation. In the second simulation, we move the 10% window by five observations, such that in our next sample, we set to zero again the earnings of a number of earners that correspond to 10 percent of our sample, while retaining the observed earnings of the five lowest paid mothers. In the third simulation, we move again the 10% window by five observations, and so forth. We repeat this procedure 85 times until the 10% window reaches the very top of the earnings distribution. As such, in the sample used to compute the last simulation, the top earners (corresponding to 10 percent of the sample) have their earnings set to zero.

The results are presented in Fig. A.3. The figure shows the aggregate increase in income taxes and the aggregate decrease in credits and transfers for both level of governments (provincial: Québec and federal: Canada). The x-axis represents the simulation number, with simulation zero being the one which excludes mothers with the lowest earnings and simulation eighty-five being the one which excludes the highest earning mothers. At the provincial level (Québec), this figure shows that the transfers to families decrease due to the increased participation rate in all simulations. The amounts range from approximately −10 to −50 million dollars. The decrease is however not monotonic. Wealthy mothers at the provincial level are in families that receive fewer credits and transfers if they do not work. The decrease at the federal level (Canada) is however monotonic. The income threshold for credits and transfers is much higher at the federal level. The amounts range from approximately −40 million to almost −120 million dollars. Therefore, the credits decreased between 50 and approximately 170 million dollars when we combine both level of governments.

In terms of income taxes, the increase at the provincial level (also shown in Fig. 3) starts at about 50 million and increases more or less monotonically up to 450 million, and from 50 to 550 million at the federal level. The overall gains from income taxes range from 100 to 1000 million dollars. The total benefits to the government (including increased income taxes and reduced credits and transfers) are between 60 and 500 million dollars at the provincial level, and 90–670 million dollars at the federal level. As such, the combined total benefits range between 150 million and 1.2 billion dollars.

For fiscal year 2004–2005, the fiscal credits devoted to the program were 1.4 billion dollars. For the same year, families paid to daycare

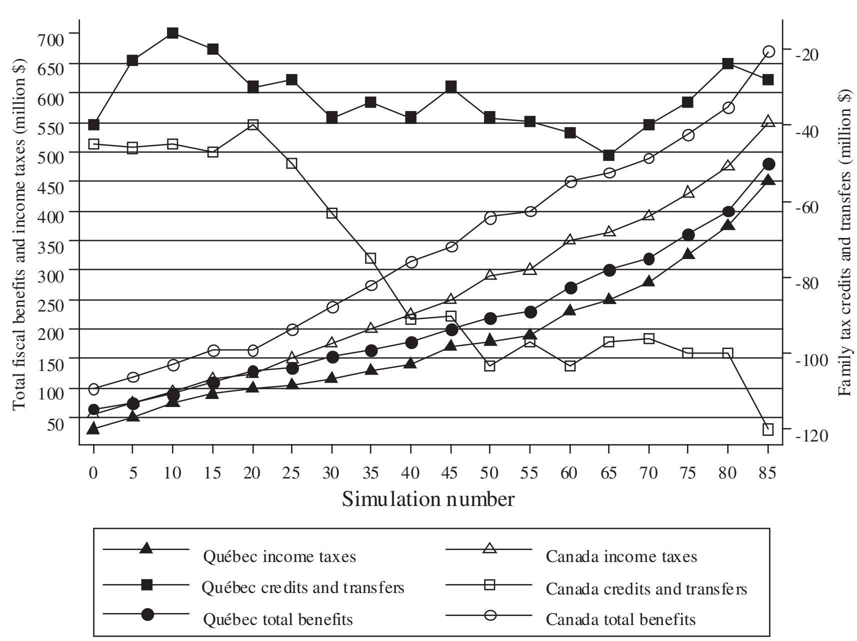

Fig. A.3. Simulated changes in families' credits, transfers and income taxes by level of government for year 2004. Note: Shows the simulated changes in provincial (Québec) and federal (Canada) income taxes (left axis) and credits and transfers (right axis) from a 10 percentage point increase in the labour market participation of mothers. The simulation uses SLID data for mothers with at least one child aged 1–11. Simulation number 1 assumes that the lowest earners join the labour market, while simulation number 85 assumes that the highest earners join the labour market (corresponding to 10 percent increase in labour force participation of mothers with at least on child aged 1–11). From simulation 1 to 85, the 10% window moves from the bottom of the income distribution to the top by 5 observations at each simulation.

Table A.1
Regulated childcare spaces by province (estimates) in 2001 and 2006.

	Number of regulated spaces[1]		Number of subsidised spaces[2]	Provincial subsidy in million $[3]	Median daily fee[4] infant-child	Net income threshold for subsidy[5]	Number of children 0–5 years old
Province/Year	2001	2006	2006	2006	2006	2006	2005
Newfoundland Labrador	3632	5017	1459	12,3	21–45	37,600	28,900
Prince Edward Island	3697	3394	843	4,7	20–30	51,040	7900
Nova Scotia	11,314	13,093	2804	23,7	23–27	34,992	50,900
New Brunswick	5820	13,163	3868	22,5	22–25	24,180	43,800
Québec	132,545	200,005	196,618	1493	7	N/A	434,800
Ontario	118,110	158,727	109,813	534,1	25–36	Multiple[6]	822,000
Manitoba	14,130	19,473	10,830	86,3	10–25	40,475	76,900
Saskatchewan	4106	7805	3672	22,8	13–18	54,960	67,100
Alberta	41,011	47,587	11,932	72,5	18–26	77,400	228,400
British Columbia	36,383	54,007	10,665	176,1	25–34	71,016	233,200
Canada	370,748	522,371	352,504	2448	-	-	1,993,900

1. Center-and family-based full-and part-day chilcare for preschool-aged children (0–5, and 0–4 in Québec).
2. Number of children in regulated chilcare receiving subsidies.
3. Provincial allocation (fee subsidy + one time funding + recurring funding) for regulated chilcare in 2005–2006.
4. Median daily fee (infant -older children). For Ontario, Manitoba, Saskatchewan, Alberta and BC we divided the monthly fees by 22.
5. Break-even point of eligibility for fee subsidy (net income 2005/06), 2 parents, 1 or 2 children.
6. The subsidy in Ontario depends on the cost of childcare amd the net income of the family.
Source: Friendly et al. (2009).

providers approximately $345 million dollars (189,380 subsidised spaces times 52 weeks times $35 per week). Families supported around 20% of the total cost (1745 millions). For year 2010–2011, this ratio decreased to 15.6%. As such, at the Québec government level, even in the best case scenario, the costs were larger than the benefits (1.7 billion versus 500 million). When we look at both levels of government, and again assume the best case scenario, the costs are only slightly larger than the fiscal benefits (1.7 billion dollars versus 1.2 billion). For more realistic cases, the gains are likely to be relatively small compared to the costs. For example, if it is the median wage earners who are enticed by the policy to enter the

Table A.2
Estimated effects of the policy on weekly hours spent in the primary mode of care by the mother's level of education.

Age of children	1	2	3	4	5 not eligible for K-5	5 eligible for K-5	1–4
Hours of care							
Some postsecondary or less							
β_3 (1998–99)	1.10	6.65***	−1.62	1.55	1.43	−6.63***	2.11*
	(1.41)	(2.53)	(2.26)	(2.41)	(2.65)	(1.40)	(1.16)
β_4 (2000–01)	1.10	7.35***	3.34*	5.84**	4.37	−6.00***	4.60***
	(1.89)	(1.92)	(1.79)	(2.81)	(3.13)	(1.79)	(1.17)
β_5 (2002–03)	3.22	9.75***	6.47**	5.01**	11.25***	−5.86***	6.36***
	(2.46)	(2.72)	(2.52)	(2.22)	(2.64)	(1.5)	(1.24)
β_6 (2004–05)	10.32***	12.90***	8.65***	10.34***	3.28	−3.42*	10.42***
	(2.89)	(2.45)	(2.67)	(3.11)	(4.11)	(1.78)	(1.44)
β_7 (2006–07)	8.99***	13.62***	9.69***	14.95***	23.48***	−6.61***	12.30***
	(2.63)	(2.74)	(2.63)	(2.75)	(6.49)	(1.90)	(1.39)
β_8 (2008–09)	6.41***	9.40**	9.05***	15.75***	7.08*	−5.48***	9.60***
	(2.39)	(3.88)	(2.58)	(3.52)	(4.29)	(1.90)	(1.63)
N	10,060	7158	9238	7068	3525	8519	33,524
Postsecondary diploma or more							
β_3 (1998–99)	4.36**	−1.70	3.49	4.17	2.43	−3.21	2.33*
	(1.93)	(2.53)	(2.55)	(2.90)	(4.36)	(2.01)	(1.28)
β_4 (2000–01)	7.60***	8.48***	7.56***	3.05	8.65*	−2.25	6.57***
	(2.45)	(2.35)	(1.94)	(2.78)	(4.85)	(2.23)	(1.25)
β_5 (2002–03)	10.94***	10.42***	10.33***	11.25***	11.95***	−2.51	10.71***
	(2.63)	(2.55)	(2.49)	(2.59)	(4.33)	(2.23)	(1.38)
β_6 (2004–05)	10.97***	10.59***	10.28***	12.33***	11.33*	−5.47**	11.02***
	(2.28)	(3.02)	(2.68)	(3.05)	(6.34)	(2.19)	(1.44)
β_7 (2006–07)	6.46***	7.23***	14.94***	12.92***	11.01**	−1.66	10.18***
	(2.47)	(2.53)	(2.06)	(2.75)	(4.73)	(2.18)	(1.30)
β_8 (2008–09)	6.69***	8.76***	13.85***	13.98***	22.53***	−1.78	10.89***
	(2.35)	(2.63)	(2.15)	(2.49)	(4.75)	(2.15)	(1.26)
N	9303	6820	8867	6734	3069	7905	31,724
Conditional hours of care (intensity)							
Some postsecondary or less							
β_3 (1998–99)	−3.64*	2.11	−0.63	7.95***	0.30	−10.38***	1.64
	(1.95)	(3.23)	(2.94)	(2.96)	(6.36)	(2.08)	(1.52)
β_4 (2000–01)	0.47	1.97	2.29	7.02**	−2.41	−9.89***	3.22**
	(2.64)	(2.88)	(1.99)	(2.94)	(6.83)	(2.61)	(1.43)
β_5 (2002–03)	−4.21	7.13***	4.03	4.82*	4.61	−10.89***	3.45**
	(2.93)	(2.71)	(2.72)	(2.56)	(6.23)	(2.12)	(1.44)
β_6 (2004–05)	3.85	5.82**	4.00	10.21***	9.91	−7.48***	5.67***

(continued on next page)

Table A.2 (*continued*)

Age of children	1	2	3	4	5 not eligible for K-5	5 eligible for K-5	1–4
	(2.97)	(2.70)	(2.85)	(3.36)	(6.89)	(2.35)	(1.59)
β_7 (2006–07)	1.12	3.27	3.71	11.73***	15.39**	−14.44***	5.44***
	(2.43)	(3.03)	(2.46)	(2.77)	(7.22)	(2.73)	(1.38)
β_8 (2008–09)	−3.25	5.99*	2.80	11.80***	6.78	−13.02***	3.57**
	(2.74)	(3.60)	(2.61)	(3.75)	(7.29)	(3.00)	(1.71)
N	4057	3094	4289	3187	1449	3493	14,627
Postsecondary diploma or more							
β_3 (1998–99)	1.61	−1.48	4.74**	5.42**	4.22	−8.15***	2.22*
	(1.73)	(2.61)	(2.32)	(2.45)	(3.87)	(2.20)	(1.25)
β_4 (2000–01)	4.24**	4.19*	3.61*	−1.16	7.21	−9.61***	2.55**
	(1.89)	(2.18)	(1.97)	(2.61)	(4.75)	(2.35)	(1.11)
β_5 (2002–03)	5.16**	5.85**	5.86***	4.60**	8.30**	−8.68***	5.05***
	(2.12)	(2.38)	(2.21)	(2.30)	(3.90)	(2.43)	(1.19)
β_6 (2004–05)	4.49**	6.35**	4.03	5.53**	13.23**	−11.95***	4.71***
	(1.87)	(2.50)	(2.46)	(2.45)	(5.14)	(2.42)	(1.25)
β_7 (2006–07)	0.22	0.86	6.44***	7.21***	3.80	−11.35***	3.50***
	(2.06)	(2.37)	(2.11)	(2.2)	(4.45)	(2.38)	(1.12)
β_8 (2008–09)	1.73	1.67	7.36***	4.29*	16.06***	−10.34***	3.67***
	(2.00)	(2.42)	(2.08)	(2.42)	(4.36)	(2.34)	(1.17)
N	5548	4304	5776	4256	1803	4512	19,884

Note: The dependent variables are the number of hours in care for children whose mother has some postsecondary education or less (panel 1) and at least a postsecondary diploma (panel 2), and the conditional number of hours in care for children whose mother has some postsecondary education or less (panel 3) and at least a postsecondary diploma (panel 4). Children from all family types are included. We include the same controls as in Table 3. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is p < 0.01, ** is p < 0.05, and * is p <0.1.

labour market, then the fiscal gains for the provincial government are less than 200 million dollars. However, since 2005, the family credits for low-income families in Québec are higher, therefore the aggregate savings from transfers are probably higher as of 2005.

Other possible gains could be accounted for, such as those from inducing mothers on welfare to move into the labour market, and those associated with getting childcare workers to report their earnings, but these would be fairly small as individuals on welfare typically do not earn high income when they return to work and daycare workers do not pay considerable amounts of taxes. As such, even then the policy seems very costly. Furthermore, previous research documented harmful effects of the childcare policy on early childhood developmental measures (before age 5 years old). We pursue this line of research by investigating whether the effects have persisted over time (up to 2009) and whether they persist up to age 5 years old at the time of school entry using additional developmental measures.

Table A.3
Estimated effects of the policy on cumulative hours of care 1.

Cumulative hours of care	1- to 5-year-olds not eligible for K-5
β_4 (2000–01)	4.83*
	(2.68)
β_5 (2002–03)	8.82***
	(2.24)
β_6 (2004–05)	7.29**
	(3.14)
β_7 (2006–07)	12.75***
	(2.89)
β_8 (2008–09)	16.99***
	(2.87)
Observations	4171

Note: The dependent variable is the cumulative number of hours in care (including zero hours of care) between age 1–5 years old for longitudinal children from waves 1 to 8. Children from all family types are included. We include the same controls as in Table 3. Standard errors are in parentheses and all are bootstrapped to account for the sampling design of the NLSCY. Statistical significance is denoted using asterisks: *** is p < 0.01, ** is p < 0.05, and * is p < 0.1.

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